

Probability and random variables

Probability

Sample space, probability and probability space.

- A random expt is any situation which has a set of possible outcomes, each of which occurs w/ a particular probability
- To mathematically describe a random expt, we must specify:
 - ↳ 1) The sample space Ω : The set of all possible outcomes of the random expt.
We can call any subset $A \subseteq \Omega$ an event
 - ↳ 2) The probability P : A mapping/function from events to a no. in the interval $[0,1]$
- We call (Ω, P) the probability space.

Axioms of probability

- Probability theory is based on the Kolmogorov axioms:

↳ Axiom I: The probability of an event is a nonnegative real no.

$$P(A) \in \mathbb{R}, P(A) \geq 0 \quad \forall A \subseteq \Omega$$

↳ Axiom II: The sample space / certain event has unit probability

$$P(\Omega) = 1.$$

↳ Axiom III: Additivity for incompatible events (disjoint sets)

$$P(A \cup B) = P(A) + P(B) \quad \text{if } A \cap B = \emptyset.$$

- These three axioms are sufficient to prove the following:

↳ Monotonicity:

$$\text{if } A \subseteq B, \text{ then } P(A) \leq P(B)$$

↳ Probability of the empty set:

$$P(\emptyset) = 0$$

↳ Complement rule:

$$P(A') = 1 - P(A)$$

↳ Numeric bound:

$$0 \leq P(A) \leq 1 \quad \forall A \subseteq \Omega$$

↳ Addition law:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

↳ Sum rule:

$$P(A) = P(A \cap B) + P(A \cap B')$$

↳ Additivity extended: $P(A) = \sum_{\omega \in A} P(\omega)$ where $\{\omega \in \Omega\}$ are mutually-disjoint individual outcomes.

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Defining probability

- Define the indicator function for a set or event E

$$I_E(t) = \begin{cases} 0 & t \in E \\ 1 & t \notin E \end{cases}$$

- For a finite discrete set $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$, let $p_1, p_2, \dots, p_n$ be non-negative no. that sum to 1. For any event E , set

$$P(E) = \sum_{i=1}^n I_E(\omega_i) p_i$$

then P satisfies the axioms.

- For an infinite discrete set $\Omega = \{\omega_1, \omega_2, \dots\}$, let $p_1, p_2, \dots$ be a non-negative sequence that sum to 1. For any event E , set

$$P(E) = \sum_{i=1}^{\infty} I_E(\omega_i) p_i$$

then P satisfies the axioms.

- For the set of real no $\Omega = \mathbb{R}$, we define a pdf $f(t)$ that satisfies

$$f(t) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} f(t) dt = 1$$

For any event E , set

$$P(E) = \int_{-\infty}^{\infty} I_E(t) f(t) dt$$

then P satisfies the axioms.

Conditional probability

- The conditional probability of event A occurring given that event B has occurred is defined to be.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{if } P(B) > 0$$

- From this, we can get the multiplication rule

$$P(A \cap B) = P(A|B) \cdot P(B)$$

In general, we can define the probability chain rule

$$P(A_1 \dots A_n) = P(A_1) \prod_{i=2}^n P(A_i | A_1 \dots A_{i-1})$$

- Applying the sum rule, we get the law of total probability

$$P(A) = P(A \cap B) + P(A \cap B')$$

$$= P(A|B)P(B) + P(A|B')P(B')$$

In general, for a set of pairwise incompatible events B_i , st. $\bigcup_i B_i = \Omega$,

$$P(A) = \sum_i P(A|B_i) P(B_i)$$

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Bayes' theorem

- We can write $P(A|B)$ in two ways using the multiplication rule.

$$P(A|B) = P(A|B)P(B) = P(B|A)P(A)$$

Rearranging, we get Bayes' thm. (replaces $P(A|B)$ w/ $P(B|A)$)

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

- Bayes' thm. is often used w/ the law of total probability.

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B')P(B')}$$

Independence

- Two events are independent if the knowledge that one occurred does not change the probability that the other one occurs.
- Mathematically, A and B are independent if.

$$P(A) = P(A|B) = P(A|B')$$

Using the multiplication rule, we get the formal defn of independence

$$P(A|B) = P(A)P(B)$$

Random variables

Random variables (RVs)

- Given a probability space $(\Omega, \mathcal{F}, P)$, a RV is a function $X(\omega)$ which maps each element ω of the sample space Ω onto a pt. on the real line.
- We can define a RV by listing the range of $X(\omega)$ along w/ the probability the RV takes those values.
- For any set $A \subset (-\infty, \infty)$, we define

$$P(X \in A) = P(\{\omega : X(\omega) \in A\})$$

- For a discrete RV X , we define the probability mass function (pmf) P_X to be.

$$P_X(x_i) = P(X = x_i), \text{ where } \sum_i P_X(x_i) = 1.$$

$$\text{For any set } A, P(X \in A) = \sum_i I_A(x_i) P_X(x_i)$$

- For a continuous RV X , we define the probability density function (pdf) f_X to be

$$f_X(x) = P(X \in \varepsilon), \text{ where } \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

$$\text{For any set } A, P(X \in A) = \int_{-\infty}^{\infty} I_A(x) f_X(x) dx.$$

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cumulative distribution function (cdf)

- For discrete/continuous RV, the cdf F_X is defined as

$$F_X(x) = P(X \leq x)$$

note $P(X > x) = 1 - F_X(x)$

- the cdf $F_X(x)$ has the following properties:

$$\hookrightarrow 0 \leq F_X(x) \leq 1$$

$$\hookrightarrow P(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

$\hookrightarrow F_X(x)$ is non-decreasing as x increases

$$\hookrightarrow \lim_{x \rightarrow -\infty} F_X(x) = 0 \text{ and } \lim_{x \rightarrow \infty} F_X(x) = 1$$

- For continuous RV X , $F_X(x) = \int_{-\infty}^x f_X(x) dx \rightarrow$ area under f_X is a continuous function $\rightarrow F_X$ is continuous.

- For continuous RV X , the cdf F_X and pdf f_X are related by:

$$F_X(x) = \int_{-\infty}^x f_X(x) dx \xleftrightarrow{\text{FTC}} f_X(x) = \frac{dF_X(x)}{dx} \quad \leftarrow \text{useful for finding pdf of transformed variables!}$$

- For discrete RV X w/ range $\{x_1, \dots, x_n\}$, $F_X(x) = \sum_{i=1}^n P(X_i) I_{[x_i, \infty)}(x) \rightarrow F_X(x)$ is a step function that jumps w/ each $x_i \rightarrow F_X$ is right-continuous, i.e. $F_X(x) = \lim_{x \rightarrow x^+} F_X(x) \quad \forall x$.

Expectation

- The expected value of X , $E[X]$ is defined as:

$$E[X] = \begin{cases} \sum x_i P(X_i) & [\text{discrete}] \\ \int_{-\infty}^{\infty} x f_X(x) dx & [\text{continuous}] \end{cases}$$

- The law of large no.s (LLN) states that:

$$\frac{1}{n} \sum_{i=1}^n X_i \rightarrow E[X] \text{ as } n \rightarrow \infty$$

so we can think of $E[X]$ as the empirical average of independent samples of X , X_i .

- For any function $g(x)$, its expectation $E[g(X)]$ is computed using:

$$E[g(X)] = \begin{cases} \sum g(x_i) P(X_i) & [\text{discrete}] \\ \int_{-\infty}^{\infty} g(x) f_X(x) dx & [\text{continuous}] \end{cases}$$

- For $g(x) = I_A(x)$, its expectation is the probability of event A .

$$E[I_A(x)] = \begin{cases} \sum I_A(x_i) P(X_i) \\ \int_{-\infty}^{\infty} I_A(x) f_X(x) dx \end{cases} = P(X \in A)$$

The LLN formalises the freq. interpretation of probability.

$$\frac{1}{n} \sum_{i=1}^n I_A(x_i) \rightarrow E[I_A(x)] = P(A \text{ occurs}) \text{ as } n \rightarrow \infty$$

Transformation of random variable.

- Consider a continuous RV X w/ pdf f_X and cdf F_X . Define a new RV $Y = g(X)$. For strictly increasing/decreasing g , there exists an inverse g^{-1} , and the pdf of the transformed RV, f_Y , is given by:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$$

$\hookrightarrow$ increasing g [$g(x) \leq y \Leftrightarrow x \leq g^{-1}(y)$]: $F_Y(y) = P(g(X) \leq y) = P(X \leq g^{-1}(y)) = F_X(g^{-1}(y))$

$$\therefore f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{d}{dy} F_X(g^{-1}(y)) = \frac{dF_X(g^{-1}(y))}{dx} \frac{dx}{dy} = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$$

$\hookrightarrow$ decreasing g [$g(x) \leq y \Leftrightarrow x \geq g^{-1}(y)$]: $F_Y(y) = P(g(X) \leq y) = P(X \geq g^{-1}(y)) = 1 - F_X(g^{-1}(y))$

$$\therefore f_Y(y) = \frac{dF_Y(y)}{dy} = \frac{d}{dy} (1 - F_X(g^{-1}(y))) = - \frac{dF_X(g^{-1}(y))}{dx} \frac{dx}{dy} = f_X(g^{-1}(y)) \left| \frac{dg^{-1}(y)}{dy} \right|$$

* In general, express $F_Y(y)$ in terms of F_X , then differentiate w.r.t y to get $f_Y(y)$.

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Bivariate random variables

Bivariate discrete random variables.

- consider two discrete RVs X, Y where $X \in \{x_1, \dots, x_n\}$ and $Y \in \{y_1, \dots, y_m\}$. Define the joint pmf P_{XY}

$$P_{XY}(x_i, y_j) = P(X=x_i, Y=y_j)$$

- We can derive the marginal pmfs P_X, P_Y given the joint pmf P_{XY} .

$$P_X(x_i) = \sum_{j=1}^m P_{XY}(x_i, y_j)$$

$$P_Y(y_k) = \sum_{i=1}^n P_{XY}(x_i, y_k)$$

- Two discrete RVs X and Y are independent if

$$P_{XY}(x, y) = P_X(x) P_Y(y) \quad \forall (x, y)$$

- For the discrete RVs X, Y , the conditional pmf of X given $Y=y$, $P_{X|Y}$ is defined as

$$P_{X|Y}(x|y) = \frac{P_{XY}(x, y)}{P_Y(y)}$$

Bivariate continuous random variables.

- consider two continuous RVs X, Y , we define the joint pdf f_{XY} to be a non-negative function st.

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} I_A(x) I_B(y) f_{XY}(x, y) dx dy = P(X \in A, Y \in B), \text{ where } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1.$$

- We can derive the marginal pdfs f_X, f_Y given the joint pdf f_{XY} .

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

- Two continuous RVs X and Y are independent iff

$$f_{XY}(x, y) = f_X(x) f_Y(y)$$

- For the continuous RVs X, Y , the conditional pdf of X given $Y=y$, $f_{X|Y}$ is defined as

$$f_{X|Y}(x|y) = \frac{f_{XY}(x, y)}{f_Y(y)}$$

Sum of random variables.

- Let X, Y be two independent RVs and let $Z = X + Y$.

$$\text{For discrete } X, Y, P_Z(z) = \sum_{x \in \mathcal{X}} P_X(x) P_Y(z-x) = P_X * P_Y$$

$$\begin{aligned} \hookrightarrow P_Z(z) &= \sum_{x \in \mathcal{X}} P_{X+Y}(x, z) = \sum_{x \in \mathcal{X}} P_Z(x, z) P_X(x) = \sum_{x \in \mathcal{X}} P(X+Y=z | X=x) P_X(x) = \sum_{x \in \mathcal{X}} P(Y=z-x | X=x) P_X(x) \\ &= \sum_{x \in \mathcal{X}} P_Y(z-x) P_X(x) = \sum_{x \in \mathcal{X}} P_Y(z-x) P_X(x) = P_X * P_Y. \end{aligned}$$

$$\text{For continuous } X, Y, f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x) dx = f_X * f_Y$$

$$\hookrightarrow F_Z(z) = P(Z \leq z) = P(X+Y \leq z) = \iint_{x+y \leq z} f_{XY}(x, y) dx dy = \int_{-\infty}^{\infty} \int_{-\infty}^{z-y} f_{XY}(x, y) dx dy$$

$$\text{sub } u = x+y: F_Z(z) = \int_{-\infty}^{\infty} \int_{-\infty}^z f_{XY}(u-y, y) du dy = \int_{-\infty}^{\infty} \int_{-\infty}^z f_X(u-y) f_Y(y) du dy$$

$$\text{Diff wrt } z: f_Z(z) = \frac{d}{dz} F_Z(z) = \int_{-\infty}^{\infty} f_X(z-y) f_Y(y) dy = f_X * f_Y.$$

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Expectation of bivarlates

- The expectation of a function $g(X,Y)$ of the bivariate (X,Y) is given by

$$E[g(X,Y)] = \begin{cases} \sum_{x,y} g(x,y) P_{X,Y}(x,y) & \text{[discrete]} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y) dx dy & \text{[continuous]} \end{cases}$$

The conditional expectation of $g(X,Y)$ given $Y=y$ is then

$$E[g(X,Y) | Y=y] = \begin{cases} \sum_x g(x,y) P_{X|Y}(x|y) & \text{[discrete]} \\ \int_{-\infty}^{\infty} g(x,y) f_{X|Y}(x|y) dx & \text{[continuous]} \end{cases}$$

- We can compute $E[g(X,Y)]$ by considering $E[g(X,Y) | Y=y]$. For the continuous case,

$$E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f_{X,Y}(x,y) dx dy = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} g(x,y) f_{X|Y}(x|y) dx \right) f_Y(y) dy$$

$$E[g(X,Y)] = \int_{-\infty}^{\infty} E[g(X,Y) | Y=y] f_Y(y) dy = E_Y[E[g(X,Y) | Y]]$$

Note that for a function $h(Y)$, $E[h(Y)] = \int_{-\infty}^{\infty} h(y) f_Y(y) dy$

- The above result also holds for the discrete case

Law of iterated expectation (Adam's law)

- The law of iterated expectation states that

$$E[X] = E_Y[E[X|Y]]$$

- consider the continuous case (result also holds for discrete case)

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx = \int_{-\infty}^{\infty} x \left(\int_{-\infty}^{\infty} f_{X,Y}(x,y) dy \right) dx = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x f_{X|Y}(x|y) dx \right) f_Y(y) dy = \int_{-\infty}^{\infty} E[X|Y=y] f_Y(y) dy = E_Y[E[X|Y]]$$

Multivariate random variables

Random vectors

- Let $X_1, X_2, \dots, X_n$ be n continuous RVs. We call $X = (X_1, \dots, X_n) \in \mathbb{R}^n$ a continuous random vector

(similarly we can define a discrete random vector, but we will consider the continuous case in this section)

- For a continuous random vector $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, we define the pdf f to be a non-negative function that integrates to unity which satisfies the following for all events $A_1, \dots, A_n$.

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \int_{A_1} \dots \int_{A_n} I_{A_1}(x_1) \dots I_{A_n}(x_n) f(x_1, \dots, x_n) dx_1, \dots, dx_n$$

If event $A_i = [a_i, b_i]$, then we can replace the range w/

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \int_{a_1}^{b_1} \dots \int_{a_n}^{b_n} f(x_1, \dots, x_n) dx_1, \dots, dx_n$$

- The marginal pdf of X_i can be obtained by integrating $f(x_1, \dots, x_n)$ over the full range of all variables except x_i

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, \dots, x_n) dx_1 \dots dx_{i-1} dx_{i+1} \dots dx_n$$

- The n RVs $X_1, \dots, X_n$ are independent iff for every $A_1, \dots, A_n$

$$P(X_1 \in A_1, \dots, X_n \in A_n) = P(X_1 \in A_1) \dots P(X_n \in A_n)$$

This is equivalent to checking that the joint pdf reduces to the product of marginals.

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \dots f_{X_n}(x_n)$$

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Transformation of random vectors

- Consider the transformation $Y = G(X)$, i.e.

$$\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} g_1(x_1, \dots, x_n) \\ \vdots \\ g_n(x_1, \dots, x_n) \end{bmatrix}$$

If G is invertible then $X = G^{-1}(Y)$, let $H(Y) = G^{-1}(Y)$, then

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} h_1(y_1, \dots, y_n) \\ \vdots \\ h_n(y_1, \dots, y_n) \end{bmatrix}$$

Performing a change of variable during integration, we get

$$f_Y(y) = f_X(H(y)) |\det J(y)|$$

where the Jacobian $|\det J(y)|$ is the magnitude of the determinant of the matrix of partial derivatives of $H(y)$

$$J(y) = \begin{bmatrix} \frac{\partial h_1}{\partial y_1} & \dots & \frac{\partial h_1}{\partial y_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial h_n}{\partial y_1} & \dots & \frac{\partial h_n}{\partial y_n} \end{bmatrix}$$

characteristic function.

- The characteristic function of a RV X is defined as

$$\phi_X(t) = E[\exp(itX)] \quad t \in \mathbb{R}$$

The characteristic function of a random vector $X = (X_1, \dots, X_n)$ is defined as

$$\phi_X(t) = E[\exp(it^T X)] \quad t \in \mathbb{R}^n$$

- Suppose that X and Y are RVs/random vectors w/ $\phi_X(t) = \phi_Y(t)$ for all $t \in \mathbb{R} / t \in \mathbb{R}^n$, then X and Y have the same probability distribution. (since the characteristic function uniquely describes a pdf).

- consider the RV $Y = \sum_{i=1}^n X_i$, where X_i are independent RVs. Its characteristic function $\phi_Y(t)$ is

$$\phi_Y(t) = E[\exp(itY)] = E[\exp(itX_1) \dots \exp(itX_n)] = E[\exp(itX_1)] \dots E[\exp(itX_n)] = \phi_{X_1}(t) \dots \phi_{X_n}(t)$$

i.e. the characteristic function of the sum of independent RVs is the product of the individual characteristic functions.

- we can use the characteristic function to compute moments, $E[X^n]$. consider $\frac{d^n}{dt^n} \phi_X(t)$.

$$\frac{d^n}{dt^n} \phi_X(t) = E\left[\frac{d^n}{dt^n} \exp(itX)\right] = E[i^n X^n \exp(itX)]$$

If we set $t=0$, we get $\left. \frac{d^n}{dt^n} \phi_X(t) \right|_{t=0} = i^n E[X^n]$. $\rightarrow E[X^n] = \frac{1}{i^n} \left. \frac{d^n}{dt^n} \phi_X(t) \right|_{t=0}$

The Gaussian vector.

- For a Gaussian vector $X = (X_1, \dots, X_n) \in \mathbb{R}^n$, the pdf $f(x_1, \dots, x_n)$ is given by

$$f(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-m)^T \Sigma^{-1}(x-m)\right)$$

where m is the mean vector and Σ is the covariance matrix.

$$m_i = E[X_i]$$

$$[\Sigma]_{ij} = E[(X_i - m_i)(X_j - m_j)]$$

* For independent X_i , the covariance matrix Σ is diagonal, i.e. $[\Sigma]_{ij} = 0$ for $i \neq j$.

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Transformation of the Gaussian vector

- consider a Gaussian vector $X = (X_1, \dots, X_n)$, where each X_i is independent and $X_i \sim N(0, 1)$.

Let $\underline{S}$ be an invertible matrix and $\underline{m}$ be a column vector, and define $Y = \underline{m} + \underline{S}X$.

- Using the change of variable result, where $Y = g(X) = \underline{m} + \underline{S}X \rightarrow H(Y) = g^{-1}(Y) = \underline{S}^{-1}(Y - \underline{m}) \rightarrow J(Y) = \underline{S}^{-1}$.

$$f_Y(Y) = f_X(\underline{S}^{-1}(Y - \underline{m})) |\det \underline{S}^{-1}|$$

where $f_X(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2}} \exp(-\frac{1}{2} \underline{x}^T \underline{x})$.

- We can therefore rewrite $f_Y(Y)$ as

$$f_Y(Y) = \frac{|\det \underline{S}^{-1}|}{(2\pi)^{n/2}} \exp\left[-\frac{1}{2} (Y - \underline{m})^T (\underline{S}^{-1})^T \underline{S}^{-1} (Y - \underline{m})\right]$$

i.e. Y is a Gaussian vector w/ mean $\underline{m}$ and covariance matrix $\underline{S} \underline{S}^T$

(Note that $|\det \underline{S}^{-1}| = \frac{1}{|\det \underline{S}|}$ and $|\det \underline{S} \underline{S}^T|^{1/2} = |\det \underline{S} \det \underline{S}^T|^{1/2} = |\det \underline{S}|$).

- In general, an affine transformation (shift + linear stretch) of a Gaussian vector is a Gaussian vector.

We can use this result to generate any Gaussian vector $N(\underline{m}, \underline{\Sigma})$.

↳ decompose the symmetric matrix $\underline{\Sigma} = \underline{S} \underline{S}^T$ (use Cholesky decomposition)

↳ Output $\underline{m} + \underline{S}X$, where $X = (X_1, \dots, X_n)$ and each X_i is independent w/ $X_i \sim N(0, 1)$.

Characteristic function of a Gaussian

- consider a Gaussian RV X , where $X \sim N(\underline{m}, \sigma^2)$. Its characteristic function $\phi_X(t)$ is

$$E[\exp(itX)] = \int_{-\infty}^{\infty} e^{itx} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2\sigma^2}(x-\underline{m})^2\right) dx \stackrel{s=x-\underline{m}}{=} e^{it\underline{m}} \int_{-\infty}^{\infty} e^{it\underline{s}} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{s^2}{2\sigma^2}\right) ds = e^{it\underline{m}} e^{-\frac{1}{2}\sigma^2 t^2}$$

$$\text{since } FT[e^{-\frac{1}{2}\sigma^2 t^2}] = \frac{\sqrt{\pi}}{\sigma/\sqrt{2}} e^{-\frac{s^2}{4(\sigma/2)^2}} = \frac{\sqrt{\pi}}{\sigma} e^{-\frac{s^2}{2\sigma^2}} \rightarrow \text{IFT}\left[\frac{\sqrt{\pi}}{\sigma} e^{-\frac{s^2}{2\sigma^2}}\right] = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\sqrt{\pi}}{\sigma} e^{-\frac{s^2}{2\sigma^2}} e^{it\underline{s}} ds = e^{-\frac{1}{2}\sigma^2 t^2}$$

- Consider the transformation of a Gaussian vector $X = (X_1, \dots, X_n)$, $X_i \text{ iid } \sim N(0, 1)$ and $Y = \underline{m} + \underline{S}X$.

Noting that $t^T Y = t^T \underline{m} + t^T \underline{S}X = t^T \underline{m} + \underline{r}^T X$, where we define $\underline{r} = \underline{t}^T \underline{S}$

$$\phi_Y(t) = E[\exp(it^T Y)] = E[\exp(it^T \underline{m}) \exp(\underline{r}^T X)] = E[\exp(it^T \underline{m}) \exp(ir_1 X_1) \dots \exp(ir_n X_n)]$$

$$= \exp(it^T \underline{m}) E[\exp(ir_1 X_1)] \dots E[\exp(ir_n X_n)] = \exp(it^T \underline{m}) \exp(-\frac{1}{2} r_1^2) \dots \exp(-\frac{1}{2} r_n^2)$$

$$= \exp(it^T \underline{m}) \exp(-\frac{1}{2} \sum_{i=1}^n r_i^2) = \exp(it^T \underline{m}) \exp(-\frac{1}{2} \underline{r}^T \underline{r}) = \exp(it^T \underline{m}) \exp(-\frac{1}{2} \underline{t}^T \underline{S} \underline{S}^T \underline{t})$$

i.e. the characteristic function of a multivariate Gaussian w/ mean $\underline{m}$ and covariance $\underline{\Sigma} = \underline{S} \underline{S}^T$.

Random process

Random process

Random process

- A discrete time random process is defined as an ensemble of functions

$$\{X_n(\omega)\}, \quad n \in (-\infty, \infty)$$

where ω is a RV w/ pdf $f_\omega(\omega)$.

- We can consider a generative model for the waveforms. We first draw a random value $\tilde{\omega}$ from the density $f_\omega(\omega)$. Then, the waveform for this value $\omega = \tilde{\omega}$ is given by $X_n(\tilde{\omega})$, $n \in (-\infty, \infty)$.
- The ensemble is built up by considering all possible values ω and their corresponding waveforms $X_n(\omega)$. The rel. freq. of each waveform $X_n(\omega)$ is determined by $f_\omega(\omega)$.
- usually, for simplicity, we can consider the random process as an infinite collection of RVs, e.g.,

$$\{X_n\}_{n=-\infty}^{\infty} = \{\dots, X_{-1}, X_0, X_1, \dots\} \quad \text{or} \quad \{X_n\}_{n=0}^{\infty} = \{X_0, X_1, \dots\}$$

- To completely specify the random process, we must specify

→ For continuous RVs $X_0, X_1, \dots$, its n th order joint pdf

$$f_{X_0, X_1, \dots, X_n}(x_0, x_1, \dots, x_n) \quad \forall n \in \mathbb{Z}^+$$

→ For discrete RVs $X_0, X_1, \dots$, its n th order joint pmf.

$$P_{X_0, X_1, \dots, X_n}(x_0, x_1, \dots, x_n) \quad \forall n \in \mathbb{Z}^+$$

correlation and covariance functions.

- The autocorrelation function of a random process $\{X_n\}$ is

$$r_{xx}[n, m] = E[X_n X_m]$$

The autocovariance function of a random process $\{X_n\}$ is

$$c_{xx}[n, m] = E[(X_n - \mu_{X_n})(X_m - \mu_{X_m})]$$

- The crosscorrelation function b/w random processes $\{X_n\}$ and $\{Y_n\}$ is

$$r_{xy}[n, m] = E[X_n Y_m]$$

The crosscovariance function b/w random processes $\{X_n\}$ and $\{Y_n\}$ is

$$c_{xy}[n, m] = E[(X_n - \mu_{X_n})(Y_m - \mu_{Y_m})]$$

strict-sense stationarity (SSS)

- A discrete-time random process $\{X_n\}$ is SSS if, for any finite k, m and $(\alpha_0, \alpha_1, \dots, \alpha_k)$,

$$f_{X_0, X_1, \dots, X_k}(\alpha_0, \alpha_1, \dots, \alpha_k) = f_{X_0, X_1, \dots, X_k}(\alpha_0, \alpha_1, \dots, \alpha_k)$$

i.e. it has the same statistical characteristics irrespective of shifts along the time axis,

+ If $X_0, X_1, \dots$ are discrete, we would use the joint pmf instead of the joint pdf to define SSS.

Wide sense stationarity (WSS)

- A discrete-time random process $\{X_n\}$ is WSS if all the following are satisfied

(i) $E[X_n] = \mu \quad \forall n$ (const. mean)

(ii) $\text{Var}[X_n] < \infty / E[X_n^2] < \infty \quad \forall n$ (finite variance/power)

(iii) $r_{XX}[n, m] \rightarrow r_{XX}[m-n] \quad (r_{XX} = r_{XX}(m-n))$

- Two discrete-time random processes $\{X_n\}, \{Y_n\}$ are jointly WSS if the following are satisfied:

(i) $E[X_n] = \mu_X, E[Y_n] = \mu_Y \quad \forall n$ (const. mean)

(ii) $\text{Var}[X_n] < \infty / E[X_n^2] < \infty, \text{Var}[Y_n] < \infty / E[Y_n^2] < \infty, \quad \forall n$ (finite variance/power)

(iii) $r_{XX}[n, m] \rightarrow r_{XX}[m-n], r_{YY}[n, m] \rightarrow r_{YY}[m-n], r_{XY}[n, m] \rightarrow r_{XY}[m-n] \quad (r = r(m-n))$

- For WSS $\{X_n\}$, $r_{XX}[k] = r_{XX}^*[k]$, i.e. $r_{XX}[k]$ is even

$$r_{XX}[k] = E[X_n X_{n+k}] = E[X_{n+k} X_n] = r_{XX}^*[k]$$

and $\max |r_{XX}[k]| = r_{XX}[0]$

$$0 \leq E[(X_{n+k} - aX_n)^2] = E[X_{n+k}^2] - 2aE[X_{n+k}X_n] + a^2E[X_n^2] = r_{XX}[0](1+a^2) - r_{XX}[k](2a)$$

$$\therefore r_{XX}[0](1+a^2) \geq 2ar_{XX}[k]$$

setting $a=1 \rightarrow r_{XX}[0] \geq r_{XX}[k]$; setting $a=-1 \rightarrow r_{XX}[0] \geq -r_{XX}[k] \Rightarrow r_{XX}[0] \geq |r_{XX}[k]|$

- For jointly WSS $\{X_n\}, \{Y_n\}$, $r_{XY}[k] = r_{YX}^*[k]$

$$r_{XY}[k] = E[X_n Y_{n+k}] = E[Y_{n+k} X_n] = r_{YX}^*[k]$$

Power spectral density,

- For WSS $\{X_n\}$, the PSD function is the DTFT of the autocorrelation function $r_{XX}[k]$.

$$S_X(f) = \sum_{k=-\infty}^{\infty} r_{XX}[k] e^{j2\pi f k}$$

$$r_{XX}[k] = \int_{-1/2}^{1/2} S_X(f) e^{-j2\pi f k} df$$

- For jointly WSS $\{X_n\}, \{Y_n\}$, the cross PSD function is the DTFT of the crosscorrelation function $r_{XY}[k]$

$$S_{XY}(f) = \sum_{k=-\infty}^{\infty} r_{XY}[k] e^{j2\pi f k}$$

$$r_{XY}[k] = \int_{-1/2}^{1/2} S_{XY}(f) e^{-j2\pi f k} df$$

- The PSD $S_X(f)$ is periodic w/ period 1

$$S_X(f+1) = \sum_{k=-\infty}^{\infty} r_{XX}[k] e^{j2\pi(f+1)k} = \sum_{k=-\infty}^{\infty} r_{XX}[k] e^{j2\pi f k} e^{j2\pi k} = S_X(f)$$

- The PSD $S_X(f)$ is real-valued and even. It is also continuous if $\sum_{k=-\infty}^{\infty} |r_{XX}[k]| < \infty$.

$$S_X(f) = \sum_{k=-\infty}^{\infty} r_{XX}[k] e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} r_{XX}[k] \cos(2\pi f k) - j \sum_{k=-\infty}^{\infty} r_{XX}[k] \sin(2\pi f k) = \sum_{k=-\infty}^{\infty} \underbrace{r_{XX}[k]}_{\text{even}} \underbrace{\cos(2\pi f k)}_{\text{even}}$$

$\therefore S_X(f)$ is the sum of real-valued, even and continuous functions $\rightarrow S_X(f)$ is real, even and continuous.

- The PSD $S_X(f)$ can be regarded as the density of power — integrating gives power,

so it must be non-negative.

- The cross PSD $S_{XY}(f)$ has the symmetry $S_{XY}(f) = S_{YX}^*(f)$

$$S_{XY}(f) = \sum_{k=-\infty}^{\infty} r_{XY}[k] e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} r_{YX}^*[k] e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} r_{YX}^*[k] e^{+j2\pi f k} = \left[\sum_{k=-\infty}^{\infty} r_{YX}[k] e^{-j2\pi f k} \right]^* = S_{YX}^*(f)$$

- The cross PSD $S_{XY}(f)$ can be regarded as a measure of the coherence between the two processes for a certain frequency.

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Linear systems and random processes.

- If the i/p $\{x_n\}$ of a LTI system w/ impulse resp $\{h_n\}$ is WSS, and $\sum_{n=-\infty}^{\infty} |h_n| < \infty$, then the o/p $\{y_n\}$ is also WSS.

(i) consider the mean $E[y_n]$.

$$E[y_n] = E\left[\sum_{k=-\infty}^{\infty} h_k x_{n-k}\right] = \sum_{k=-\infty}^{\infty} h_k E[x_{n-k}] = E[x_0] \sum_{k=-\infty}^{\infty} h_k \rightarrow \text{const. over all } n.$$

(ii) consider the autocorrelation $r_{yy}[k]$.

$$\begin{aligned} r_{yy}[k] &= E\left[\left(\sum_{i=-\infty}^{\infty} h_i x_{n-i}\right)\left(\sum_{j=-\infty}^{\infty} h_j x_{n+k-j}\right)\right] = E\left[\sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_i h_j x_{n-i} x_{n+k-j}\right] \\ &= \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_i h_j E[x_{n-i} x_{n+k-j}] = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_i h_j r_{xx}[k+i-j] = \sum_{i=-\infty}^{\infty} h_i \left(\sum_{j=-\infty}^{\infty} h_j r_{xx}[k+i-j]\right) \\ &= \sum_{i=-\infty}^{\infty} h_i (h_{k+i} r_{xx}[k+i]) = \tilde{h}_k * h_k * r_{xx}[k] \rightarrow \text{only a function of } k. \end{aligned}$$

where $\tilde{h}_k$ is the time-reversed sequence of h_k , i.e. $\tilde{h}_k = h_{-k}$.

$$(\text{Note } a_n * b_n = \sum_{k=-\infty}^{\infty} a_k b_{n-k}, \quad \tilde{a}_n * b_n = \sum_{k=-\infty}^{\infty} \tilde{a}_k b_{n-k} = \sum_{k=-\infty}^{\infty} a_{-k} b_{n-k} = \sum_{k=-\infty}^{\infty} a_k b_{n+k}).$$

(iii) consider the power $E[y_n^2] = r_{yy}[0]$.

$$r_{yy}[0] = \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_i h_j r_{xx}[i-j] \leq \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} |h_i h_j| r_{xx}[0] = r_{xx}[0] \sum_{i=-\infty}^{\infty} |h_i| \sum_{j=-\infty}^{\infty} |h_j|$$

This is finite if $\sum_{i=-\infty}^{\infty} |h_i| < \infty$.

- The crosscorrelation $r_{xy}[k]$ and o/p autocorrelation $r_{yy}[k]$ can be expressed as convolutions.

$$r_{xy}[k] = h_k * r_{xx}[k]$$

$$r_{yy}[k] = \tilde{h}_k * h_k * r_{xx}[k].$$

$$\hookrightarrow r_{xy}[k] = E[x_n y_{n+k}] = E\left[x_n \sum_{i=-\infty}^{\infty} h_i x_{n+k-i}\right] = \sum_{i=-\infty}^{\infty} h_i E[x_n x_{n+k-i}] = \sum_{i=-\infty}^{\infty} h_i r_{xx}[k-i] = h_k * r_{xx}[k].$$

$\hookrightarrow$ Derivation for $r_{yy}[k] = \tilde{h}_k * h_k * r_{xx}[k]$ found above.

$$\xrightarrow{r_{xx}[k]} \xrightarrow{h_k} \xrightarrow{r_{xy}[k]} \xrightarrow{\tilde{h}_k} r_{yy}[k]$$

- Taking the DTFT of the convolution form of the o/p autocorrelation $r_{yy}[k]$,

$$\text{DTFT}[r_{yy}[k] = \tilde{h}_k * h_k * r_{xx}[k]] \rightarrow S_y(f) = H^*(f) H(f) S_x(f) = |H(f)|^2 S_x(f)$$

$$\text{where } \text{DTFT}[\tilde{h}_k] = \sum_{k=-\infty}^{\infty} \tilde{h}_k e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} h_{-k} e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} h_k e^{-j2\pi f k} = \left(\sum_{k=-\infty}^{\infty} h_k e^{j2\pi f k}\right)^* = H^*(f).$$

Physical interpretation of PSD

- For a deterministic signal x_n , the instantaneous power is x_n^2 and the average power is $\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x_n^2$.

- For a random process $\{x_n\}$, the expected instantaneous power is $E[x_n^2]$ and the expected average

$$\text{power is } E\left[\lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x_n^2\right] = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N E[x_n^2].$$

- If $\{x_n\}$ is WSS, then $E[x_n^2] = r_{xx}[0]$ for all n , so the expected average power is $r_{xx}[0]$.

$$r_{xx}[0] = \int_{-1/2}^{1/2} S_x(f) e^{j2\pi f(0)} df = \int_{-1/2}^{1/2} S_x(f) df$$

i.e. the area under $S_x(f)$ from $[-1/2, 1/2]$ gives the total power of $\{x_n\}$.

- To find the power of a WSS process $\{x_n\}$ in a particular freq. band, say $[f_1, f_2]$ s.t. $0 < f_1 < f_2 < 1/2$, consider passing $\{x_n\}$ through an ideal bandpass filter w/ TF $H(f) = \begin{cases} 1 & \text{if } f_1 \leq f \leq f_2 \\ 0 & \text{o.w.} \end{cases}$ then find the o/p power $r_{yy}[0]$.

$$r_{yy}[0] = \int_{-1/2}^{1/2} S_y(f) df = \int_{-1/2}^{1/2} |H(f)|^2 S_x(f) df = \int_{f_1}^{f_2} S_x(f) df + \int_{f_1}^{f_2} S_x(f) df = 2 \int_{f_1}^{f_2} S_x(f) df.$$

$\rightarrow$ The power of $\{x_n\}$ in a particular freq. band $f_1 \leq f \leq f_2$ is $2 \int_{f_1}^{f_2} S_x(f) df$

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Ergodic random process.

- For an ergodic random process, we can estimate expectations by performing time-averaging on a single sample function (rather than averaging over the whole ensemble).

(i) mean ergodic $\mu = E[X_n] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} X_n \approx \frac{1}{N} \sum_{n=0}^{N-1} X_n$ ← time average

(ii) correlation ergodic $r_{xx}[k] = E[X_n X_{n+k}] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} X_n X_{n+k} \approx \frac{1}{N} \sum_{n=0}^{N-1} X_n X_{n+k}$ ← ensemble average

- A necessary and sufficient condition for mean ergodicity is given by

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=0}^{N-1} C_{xx}[k] = 0$$

A simpler sufficient condition for mean ergodicity is that

$$C_{xx}[0] < \infty$$

and

$$\lim_{N \rightarrow \infty} C_{xx}[N] = 0$$

(N/S condition true → statement true, N/S condition false → statement false)

(S condition true → statement true, S condition false → statement indeterminate)

* Unless otherwise stated, we can always assume that all signals encountered are ergodic.

Markov chains (MC)

Markov chains (MC)

limited memory → Markov property

- A random process $\{X_n\}$ is a MC if for all times $n \geq 0$,

$$P_{X_n | X_0 \dots X_{n-1}}(i_n | i_0 \dots i_{n-1}) = P_{X_n | X_{n-1}}(i_n | i_{n-1})$$

where the RVs $X_0, X_1, \dots$ can take values in the state space $S = \{1, \dots, L\}$

- The MC is completely defined by the transitional probability matrix $\underline{Q}$ and the initial distribution of the chain $\underline{\lambda}$.

→ The transitional probability matrix $\underline{Q}$ has non-negative entries and each row sums to 1.

$$\underline{Q} = \begin{bmatrix} Q_{1,1} & Q_{1,2} & \dots & Q_{1,L} \\ Q_{2,1} & Q_{2,2} & \dots & Q_{2,L} \\ \vdots & \vdots & \ddots & \vdots \\ Q_{L,1} & Q_{L,2} & \dots & Q_{L,L} \end{bmatrix}$$

where Q_{i_{n-1}, i_n} denotes the conditional prob of X_n given $(X_0, \dots, X_{n-1}) = (i_0, \dots, i_{n-1})$, i.e.

$$Q_{i_{n-1}, i_n} = P_{X_n | X_0 \dots X_{n-1}}(i_n | i_0 \dots i_{n-1})$$

→ The initial distribution $\underline{\lambda}$ has non-negative entries and they sum to 1.

$$\lambda_i = P_{X_0}(i) \quad i = 1, \dots, L$$

* For MC, we make the notation compact by dropping the subscripts, e.g. $P_{X_0 \dots X_n}(i_0 \dots i_n) \rightarrow P(i_0 \dots i_n)$

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Marginals of a Markov chain.

- The marginal of a Markov chain can be expressed as

$$P_{X_n}(i_n) = (\underline{\lambda} \underline{Q}^n)_{i_n}$$

i.e.

$$P_{X_n}(i_n) = \begin{bmatrix} 1 \times L \end{bmatrix} \begin{bmatrix} L \times L \end{bmatrix}^n = \begin{bmatrix} 1 \times L \end{bmatrix}$$

pick 1st row entry

- starting from the joint pnf $P(i_0, \dots, i_n)$,

$$P(i_0, \dots, i_n) = \frac{P(i_0, \dots, i_n)}{P(i_0, \dots, i_{n-1})} \cdot \frac{P(i_0, \dots, i_{n-1})}{P(i_0, \dots, i_{n-2})} \dots \frac{P(i_0, i_1)}{P(i_0)} P(i_0) = P(i_n | i_0, \dots, i_{n-1}) P(i_{n-1} | i_0, \dots, i_{n-2}) \dots P(i_1 | i_0) P(i_0)$$

$$P(i_0, \dots, i_n) = P(i_n | i_{n-1}) P(i_{n-1} | i_{n-2}) \dots P(i_1 | i_0) P(i_0) = Q_{i_{n-1}, i_n} Q_{i_{n-2}, i_{n-1}} \dots Q_{i_0, i_1} \lambda_{i_0}$$

Marginalising by summing over $i_0, \dots, i_{n-1}$,

$$P(i_n) = \sum_{i_0, \dots, i_{n-1} \in S} P(i_0, \dots, i_n) = \sum_{i_1, \dots, i_{n-1} \in S} Q_{i_{n-1}, i_n} \dots Q_{i_1, i_2} \sum_{i_0 \in S} Q_{i_0, i_1} \lambda_{i_0} = \sum_{i_1, \dots, i_{n-1} \in S} Q_{i_{n-1}, i_n} \dots Q_{i_1, i_2} (\underline{\lambda} \underline{Q})_{i_1}$$

$$= \sum_{i_2, \dots, i_{n-1} \in S} Q_{i_{n-1}, i_n} \dots Q_{i_2, i_3} \sum_{i_1 \in S} Q_{i_1, i_2} (\underline{\lambda} \underline{Q})_{i_1} = \sum_{i_2, \dots, i_{n-1} \in S} Q_{i_{n-1}, i_n} \dots Q_{i_2, i_3} (\underline{\lambda} \underline{Q}^2)_{i_2} = (\underline{\lambda} \underline{Q}^n)_{i_n}$$

dot product b/w $\underline{\lambda}$ and 1st col. of $\underline{Q}$

Invariant distribution of a Markov chain.

- consider a MC w/ transition probability matrix $\underline{Q}$ w/ state space S . The pnf $\underline{\pi} = (\pi_i : i \in S)$ is invariant for $\underline{Q}$ if $\forall j \in S$,

$$\pi_j = \sum_{i \in S} \pi_i Q_{i,j} = (\underline{\pi} \underline{Q})_j \quad \text{or} \quad \underline{\pi} = \underline{\pi} \underline{Q}$$

- the MC $(\underline{\pi}, \underline{Q})$ is SSS.

$$\hookrightarrow P(i_n) = (\underline{\pi} \underline{Q}^n)_{i_n} = (\underline{\pi} \underline{Q} \underline{Q}^{n-1})_{i_n} = (\underline{\pi} \underline{Q}^{n-1})_{i_n} = \dots = \pi_{i_n}$$

$$P(i_m, \dots, i_{n+k}) = Q_{i_{n+k}, i_{n+k-1}} \dots Q_{i_m, i_{m+1}} \pi_{i_m} \rightarrow \text{we have same joint pnf for any } m \rightarrow \text{SSS.}$$

- the probability distribution $\underline{\pi}$ is known as a stationary distribution.

Ergodic theorem for Markov chain.

- An irreducible MC refers to a chain where all state values in S communicate w/ each other.
- This means for any pair of states (i, j) , the MC starting at i will eventually visit j and vice versa, in a finite no. of steps.

- We can identify communicating states by drawing a graph from the transition probability matrix $\underline{Q}$.

- When the MC $(\underline{\lambda}, \underline{Q})$ w/ stationary distribution $\underline{\pi}$ is irreducible, then for any initial distribution $\underline{\lambda}$, the sample average converges to $\sum_{i \in S} \pi_i r(i)$, i.e.

$$\frac{1}{n+1} \sum_{k=0}^n r(X_k) \rightarrow \sum_{i \in S} \pi_i r(i)$$

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Time series analysis

Time series analysis.

- A time series is a set of observations $y_n, n=0,1,\dots$ arranged in increasing time.
- Typically observations are recorded at regular intervals. For start time t_0 and time interval Δ , the n th observation is recorded at $t_n = t_0 + n\Delta$.
- When given a real time-series data $y_0, y_1, \dots$, the workflow for time series analysis is
 - ↳ 1) select a probability model to represent the data
 - ↳ 2) estimate model parameters.
 - ↳ 3) deploy - simulate or forecast.
- We will consider the probability model $y_n = m_n + s_n + x_n$,
 - where m_n is the trend - evolution of mean over time
 - s_n is the seasonal component - sinusoids w/ certain period.
 - x_n is the residual - zero-mean RV (not necessarily independent over time)

Fitting the trend m_n and seasonal component s_n

- Assuming the trend to be a k th order polynomial.

$$m_n = \alpha_0 + \alpha_1 n + \dots + \alpha_k n^k.$$

We use the data $y_0 \dots y_N$ to estimate $(\alpha_0, \alpha_1, \dots, \alpha_k)$ via least squares. $\rightarrow$ so we get $\hat{m}_n$

$$\left(\frac{\partial}{\partial \alpha_0}, \frac{\partial}{\partial \alpha_1}, \dots, \frac{\partial}{\partial \alpha_k} \right) \sum_{n=0}^N (y_n - \alpha_0 - \alpha_1 n - \dots - \alpha_k n^k)^2 = (0, 0, \dots, 0)$$

- Assuming the seasonal component is in the form

$$s_n = A \cos(2\pi f n + \phi) = \alpha_1 \cos(2\pi f n) + \alpha_2 \sin(2\pi f n)$$

We use the data $y_0 \dots y_N$ to estimate (α_1, α_2) via least squares $\rightarrow$ so we get $\hat{s}_n$.

$$\left(\frac{\partial}{\partial \alpha_1}, \frac{\partial}{\partial \alpha_2} \right) \sum_{n=0}^N (\hat{y}_n - \alpha_1 \cos(2\pi f n) - \alpha_2 \sin(2\pi f n))^2 = (0, 0)$$

where $\hat{y}_n$ is the detrended data, $\hat{y}_n = y_n - m_n$.

- Guessing the trend model will result in a biased estimate $\hat{m}_n$ s.t. $E[\hat{m}_n] \neq 0$.
- Instead of using least squares, the residual is modelled using AR/MA/ARMA models.

Time series models.

Autoregressive (AR) process.

- Let $\{W_n\}_{n=-\infty}^{\infty}$ be a sequence of RVs s.t. $E[W_n] = 0 \forall n$, $E[W_i W_j] = \begin{cases} \sigma^2 & \text{for } i=j \\ 0 & \text{for } i \neq j \end{cases}$.

The AR(p) process $\{X_n\}_{n=-\infty}^{\infty}$ is defined as

$$X_n = \left(\sum_{i=1}^p a_i X_{n-i} \right) + W_n$$

where $a_1, \dots, a_p$ are const. and p is the order of the process.

- A real time-series X_n is modelled as a function of its previous values and random part.

$$X_n = \underbrace{a_1 X_{n-1} + \dots + a_p X_{n-p}}_{\text{prediction part}} + \underbrace{W_n}_{\text{random part}}$$

- Consider the AR(1) process, $X_n = aX_{n-1} + W_n$.

- (i) The AR(1) process can be expanded in terms of W_k , $k \leq n$.

$$X_n = aX_{n-1} + W_n = a(aX_{n-2} + W_{n-1}) + W_n = \sum_{k=0}^{\infty} W_{n-k} a^k$$

$$\therefore X_n = \sum_{k=0}^{\infty} W_{n-k} a^k = W_n * h_n, \text{ where } h_k = a^k$$

i.e. AR(1) is causal w/ impulse response $\{h_k\}_{k \geq 0}$ [IIR filter]

- (ii) The mean $E[X_n]$ is zero, $E[X_n] = 0$.

$$E[X_n] = E\left[\sum_{k=0}^{\infty} W_{n-k} a^k\right] = \sum_{k=0}^{\infty} a^k E[W_{n-k}] = 0$$

alternatively,

$$E[X_n] = E[aX_{n-1} + W_n]$$

$$E[X_n] = a E[X_{n-1}] + E[W_n]$$

$$E[X_n](1-a) = 0 \rightarrow E[X_n] = 0$$

- (iii) The variance $\text{Var}[X_n]$ is given by $\text{Var}[X_n] = \frac{\sigma^2}{1-a^2}$

$$\text{Var}[X_n] = E[X_n^2] - E[X_n]^2 = E\left[\left(\sum_{i=0}^{\infty} W_{n-i} a^i\right)\left(\sum_{j=0}^{\infty} W_{n-j} a^j\right)\right] = E\left[\left(\sum_{i=0}^{\infty} W_{n-i}^2 a^{2i}\right) + \text{cross terms}\right]$$

$$= \sum_{i=0}^{\infty} a^{2i} E[W_{n-i}^2] = \sum_{i=0}^{\infty} a^{2i} \sigma^2 = \frac{\sigma^2}{1-a^2}, \text{ provided } |a| < 1$$

alternatively

$$\text{Var}[X_n] = \text{Var}[aX_{n-1} + W_n]$$

$$\text{Var}[X_n] = a^2 \text{Var}[X_{n-1}] + \text{Var}[W_n]$$

$$\text{Var}[X_n](1-a^2) = \sigma^2 \rightarrow \text{Var}[X_n] = \frac{\sigma^2}{1-a^2}$$

Note $E[W_i W_j] = 0$ for $i \neq j$ $\rightarrow$ cross-terms are neglected.

- (iv) The autocorrelation $r_{xx}[k]$ is given by $r_{xx}[k] = a^k \frac{\sigma^2}{1-a^2}$

$$r_{xx}[1] = E[X_{n-1} X_n] = E[X_{n-1} (aX_{n-1} + W_n)] = aE[X_{n-1}^2] + E[X_{n-1} W_n] = a \text{Var}[X_n] = a \frac{\sigma^2}{1-a^2}$$

$$r_{xx}[2] = E[X_{n-2} X_n] = E[X_{n-2} (aX_{n-1} + W_n)] = aE[X_{n-2} X_{n-1}] + E[X_{n-2} W_n] = a r_{xx}[1] = a^2 \frac{\sigma^2}{1-a^2}$$

By inspection, we see $r_{xx}[k+1] = a r_{xx}[k]$, so $r_{xx}[k] = a^k \frac{\sigma^2}{1-a^2}$ for $k \geq 0$

- (v) The PSD $S_x(f)$ is given by $S_x(f) = \frac{\sigma^2}{1+a^2-2a\cos(2\pi f)}$

$$\begin{aligned} S_x(f) &= \sum_{k=-\infty}^{\infty} a^k \frac{\sigma^2}{1-a^2} e^{j2\pi f k} = \frac{\sigma^2}{1-a^2} \sum_{k=-\infty}^{\infty} a^k e^{j2\pi f k} = \frac{\sigma^2}{1-a^2} \left[\sum_{k=0}^{\infty} a^k e^{j2\pi f k} + \sum_{k=-\infty}^{-1} a^k e^{j2\pi f k} - 1 \right] \\ &= \frac{\sigma^2}{1-a^2} \left[\frac{1}{1-a e^{j2\pi f}} + \frac{1}{1-a e^{-j2\pi f}} - 1 \right] = \frac{\sigma^2}{1-a^2} \left[\frac{1-a e^{j2\pi f} + 1-a e^{-j2\pi f} - (1-a^2)}{(1-a e^{j2\pi f})(1-a e^{-j2\pi f})} \right] \\ &= \frac{\sigma^2}{(1-a e^{j2\pi f})(1-a e^{-j2\pi f})} = \frac{\sigma^2}{1+a^2-2a\cos(2\pi f)} \end{aligned}$$

Alternatively, the PSD $S_x(f)$ can be found using $S_x(f) = H(f)^* H(f) S_w(f)$, where $H(f) = \text{DTFT}[h_k] = \text{DTFT}[a^k]$

$$H(f) = \sum_{k=-\infty}^{\infty} h_k e^{j2\pi f k} = \sum_{k=0}^{\infty} a^k e^{j2\pi f k} = \frac{1}{1-a e^{j2\pi f}}, \quad S_w(f) = \sum_{k=-\infty}^{\infty} W_k e^{j2\pi f k} = \sigma^2 \quad (r_{ww}[k] = \begin{cases} \sigma^2 & \text{if } k=0 \\ 0 & \text{if } k \neq 0 \end{cases})$$

$$\therefore S_x(f) = H(f)^* H(f) S_w(f) = \frac{\sigma^2}{(1-a e^{j2\pi f})(1-a e^{-j2\pi f})}$$

- * (ii), (iii), (iv) $\rightarrow$ AR(1) process is WSS.

- Higher order AR models can capture finer details and short-term fluctuations.

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Moving average process (MA)

- Let $\{W_n\}_{n=-\infty}^{\infty}$ be a sequence of RVs s.t. $E[W_n] = 0 \forall n$, $E[W_n W_m] = \begin{cases} \sigma^2 & \text{for } n=m \\ 0 & \text{for } n \neq m \end{cases}$

The MA(q) process $\{X_n\}_{n=-\infty}^{\infty}$ is defined as

$$X_n = \left(\sum_{i=1}^q b_i W_{n-i} \right) + W_n$$

where $b_1, \dots, b_q$ are const. and q is the order of the process

- A real time series X_n is modelled as a function of its previous values and a random part.

$$X_n = \underbrace{b_1 W_{n-1} + \dots + b_q W_{n-q}}_{\text{predictable part}} + \underbrace{W_n}_{\text{random part}}$$

- consider the MA(q) process, $X_n = \left(\sum_{i=1}^q b_i W_{n-i} \right) + W_n$

(i) the MA(q) process can be written in terms of W_k , $k \leq n$.

$$X_n = \sum_{k=0}^q h_k W_{n-k} = W_n + h_k, \text{ where } h_k = \begin{cases} 1 & \text{if } k=0 \\ b_k & \text{if } k \in [1, q] \\ 0 & \text{o/w} \end{cases}$$

i.e. MA(q) is causal w/ impulse response $\{h_k\}_{k=0}^q$ [FIR filter]

(ii) The mean $E[X_n]$ is zero, $E[X_n] = 0$.

$$E[X_n] = E\left[\sum_{k=0}^q h_k W_{n-k}\right] = \sum_{k=0}^q h_k E[W_{n-k}] = 0.$$

(iii) The variance $\text{Var}[X_n]$ is given by $\text{Var}[X_n] = \sigma^2 (1 + b_1^2 + \dots + b_q^2)$

$$\begin{aligned} \text{Var}[X_n] &= E[X_n^2] - E[X_n]^2 = E\left[\left(\sum_{i=1}^q b_i W_{n-i} + W_n\right)\left(\sum_{j=1}^q b_j W_{n-j} + W_n\right)\right] = E\left[\sum_{i=1}^q b_i^2 W_{n-i}^2 + W_n^2 + \text{cross terms}\right] \\ &= \left(\sum_{i=1}^q b_i^2 E[W_{n-i}^2]\right) + E[W_n^2] = \left(\sum_{i=1}^q b_i^2 \sigma^2\right) + \sigma^2 = \sigma^2 (1 + b_1^2 + \dots + b_q^2) \end{aligned}$$

(iv) The autocorrelation $r_{xx}[k]$ is given by $r_{xx}[k] = \sigma^2 \sum_{i=0}^q h_i h_{i+k}$

$$r_{xx}[k] = E[X_n X_{n+k}] = E\left[\left(\sum_{i=0}^q h_i W_{n-i}\right)\left(\sum_{j=0}^q h_j W_{n+k-j}\right)\right] = E\left[\sum_{i=0}^q \sum_{j=0}^q h_i h_j W_{n-i} W_{n+k-j}\right] = \sum_{i=0}^q \sum_{j=0}^q h_i h_j E[W_{n-i} W_{n+k-j}]$$

$$E[W_{n-i} W_{n+k-j}] = \begin{cases} \sigma^2 & \text{if } i=j+k \\ 0 & \text{o/w} \end{cases} \text{ so } r_{xx}[k] = \sigma^2 \sum_{i=0}^q h_i h_{i+k}$$

Note $r_{xx}[k] = 0$ for $k > q$ (since either h_i or h_{i+k} is zero).

(v) The PSD $S_x(f)$ is given by $S_x(f) = \left(1 + \sum_{i=1}^q b_i e^{-j2\pi f i}\right) \left(1 + \sum_{i=1}^q b_i e^{j2\pi f i}\right) \sigma^2$

$$S_x(f) = |H(f)|^2 S_w(f), \text{ where } H(f) = \text{DTFT}[h_k] = \text{DTFT}[1, b_1, \dots, b_q]$$

$$H(f) = \sum_{k=0}^q h_k e^{j2\pi f k} = 1 + \sum_{k=1}^q b_k e^{j2\pi f k}, \quad S_w(f) = \sum_{k=-\infty}^{\infty} r_{ww}[k] e^{j2\pi f k} = \sigma^2 \delta[k] = \begin{cases} \sigma^2 & \text{if } k=0 \\ 0 & \text{o/w} \end{cases}$$

$$S_x(f) = H(f) H^*(f) S_w(f) = \left(1 + \sum_{k=1}^q b_k e^{j2\pi f k}\right) \left(1 + \sum_{k=1}^q b_k e^{-j2\pi f k}\right) \sigma^2$$

* (ii), (iii), (iv) $\rightarrow$ MA(q) process is WSS

Autoregressive-moving average process (ARMA).

- Let $\{W_n\}_{n=-\infty}^{\infty}$ be a sequence of RVs s.t. $E[W_n] = 0 \forall n$, $E[W_n W_m] = \begin{cases} \sigma^2 & \text{for } n=m \\ 0 & \text{for } n \neq m \end{cases}$

The ARMA(p, q) process $\{X_n\}_{n=-\infty}^{\infty}$ is defined as.

$$X_n = \sum_{i=1}^p a_i X_{n-i} + W_n + \sum_{i=1}^q b_i W_{n-i}$$

where $a_1, \dots, a_p$, $b_1, \dots, b_q$ are const. and p/q are the order of the IIR/FIR respectively.

- The ARMA(p, q) process can be written in terms of W_k , $k \leq n$.

$$X_n = \sum_{k=0}^{\infty} W_{n-k} h_k = W_n + h_k, \text{ where } h_k = \begin{cases} 1 & \text{if } k=0 \\ a_k^* + b_k & \text{if } k \in [1, q] \\ a_k^* & \text{o/w} \end{cases}$$

i.e. ARMA(p, q) is causal w/ impulse response $\{h_k\}_{k=0}^{\infty}$ [IIR filter]

- The ARMA(p, 0) process is WSS.

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AR, MA, ARMA processes as causal filters.

- AR, MA, ARMA process can act as causal filters w/ impulse response $\{h_k\}$, subject to the i/p $\{W_n\}_{n=-\infty}^{\infty}$, where $E[W_n] = 0$ and $E[W_n W_m] = \begin{cases} \sigma^2 & \text{for } n=m \\ 0 & \text{for } n \neq m \end{cases}$.

- consider the random process $\{W_n\}_{n=-\infty}^{\infty}$.

(i) Mean $E[W_n]$: $E[W_n] = 0$, i.e. const. $\forall n$.

(ii) Power $E[W_n^2]$: $E[W_n^2] = \sigma^2 < \infty$ i.e. finite power

(iii) Autocorrelation $r_{WW}[k]$: $r_{WW}[k] = E[W_n W_{n+k}] = 0 = f_W(k)$, i.e. only depends on k

$\rightarrow \{W_n\}_{n=-\infty}^{\infty}$ is WSS.

- provided for the causal filter, its impulse response $\{h_k\}$ satisfies $\sum_{k=-\infty}^{\infty} |h_k| < \infty$, then given a WSS i/p $\{X_n\}$, its o/p $\{y_n\}$ is also WSS $\rightarrow$ AR, MA, ARMA processes must be WSS.

Random phase cosine (harmonic process)

- for the random phase cosine, the amplitude a and frequency f_0 are known, but the phase ϕ is unknown and random (could correspond to an unknown delay). It has the form

$$X_n = a \cos(2\pi f_0 n + \Phi)$$

where a, f_0 are const. and $\Phi \sim \text{Unif}[0, 2\pi)$, i.e. its pdf is $P_\Phi(\phi) = \begin{cases} 1/2\pi & \text{if } \phi \in [0, 2\pi) \\ 0 & \text{o/w} \end{cases}$

* To generate $\{X_n\}$, we need to first sample Φ to get $\phi = \hat{\phi}$, then set $X_n = \cos(2\pi f_0 n + \hat{\phi})$, for $n=0, 1, \dots$

- consider the harmonic process, $X_n = a \cos(2\pi f_0 n + \Phi)$

(i) The mean $E[X_n]$ is zero, $E[X_n] = 0$

$$E[X_n] = E[a \cos(2\pi f_0 n + \Phi)] = E[a \cos(2\pi f_0 n) \cos(\Phi) - a \sin(2\pi f_0 n) \sin(\Phi)] = a \cos(2\pi f_0 n) E[\cos(\Phi)] - a \sin(2\pi f_0 n) E[\sin(\Phi)] = 0$$

since $E[\cos(\Phi)] = E[\sin(\Phi)] = 0$ for uniform distribution Φ .

(ii) the variance $\text{var}[X_n]$ is given by $\text{var}[X_n] = \frac{1}{2} a^2$

$$\text{var}[X_n] = E[X_n^2] - E[X_n]^2 = E[a^2 \cos^2(2\pi f_0 n + \Phi)] = E\left[\frac{a^2}{2} (1 + \cos(4\pi f_0 n + 2\Phi))\right] = \frac{a^2}{2} + \frac{a^2}{2} E[\cos(4\pi f_0 n + 2\Phi)] = \frac{a^2}{2}$$

(iii) the autocorrelation $r_{XX}[k]$ is given by $r_{XX}[k] = \frac{1}{2} a^2 \cos(2\pi f_0 k)$

$$\begin{aligned} r_{XX}[k] &= E[X_n X_{n+k}] = E[a \cos(2\pi f_0 n + \Phi) a \cos(2\pi f_0 (n+k) + \Phi)] = E\left[\frac{a^2}{2} (\cos(2\pi f_0 (2n+k) + 2\Phi) + \cos(2\pi f_0 k))\right] \\ &= \frac{a^2}{2} E[\cos(2\pi f_0 (2n+k) + 2\Phi)] + \frac{a^2}{2} \cos(2\pi f_0 k) = \frac{a^2}{2} \cos(2\pi f_0 k) \end{aligned}$$

(iv) the PSD $S_X(f)$ is given by $S_X(f) = \frac{\sigma^2 \pi}{2} (\delta(2\pi(f-f_0)) + \delta(2\pi(f+f_0)))$

$$S_X(f) = \sum_{k=-\infty}^{\infty} r_{XX}[k] e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} \frac{a^2}{2} \cos(2\pi f_0 k) e^{j2\pi f k} = \frac{a^2}{4} \sum_{k=-\infty}^{\infty} (e^{j2\pi(f-f_0)k} + e^{j2\pi(f+f_0)k})$$

$$= \frac{a^2}{4} [2\pi \delta(2\pi(f-f_0)) + 2\pi \delta(2\pi(f+f_0))] = \frac{a^2 \pi}{2} (\delta(2\pi(f-f_0)) + \delta(2\pi(f+f_0)))$$

* (i), (ii), (iii) $\rightarrow$ harmonic process is WSS.

- consider the inversion formula for PSD, $r_{XX}[k] = \int_{-1/2}^{1/2} S_X(f) e^{j2\pi f k} df = \int_{-1/2}^{1/2} S_X(f) \cos(2\pi f k) df + j \int_{-1/2}^{1/2} S_X(f) \sin(2\pi f k) df$

$$\therefore r_{XX}[k] = 2 \int_{-1/2}^{1/2} S_X(f) \cos(2\pi f k) df \xrightarrow{\text{divide into intervals, within } 1/k} 2 \sum_{i=1}^K S_X(f_i) \cos(2\pi f_i k) \frac{1}{2K} = \frac{1}{K} \sum_{i=1}^K S_X(f_i) \cos(2\pi f_i k)$$

i.e. we have a sum of random phase cosines w/ freq. f_i and amplitude a st. $\frac{a^2}{2} = \frac{S_X(f_i)}{K} \rightarrow a = \sqrt{\frac{2S_X(f_i)}{K}}$

$$\therefore X_n = \sum_{i=1}^K \sqrt{\frac{2S_X(f_i)}{K}} \cos(2\pi f_i n + \Phi_i) \quad \text{where } \Phi_i \text{ iid } \sim \text{Unif}[0, 2\pi)$$

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white noise process,

- A WSS process $\{x_n\}$ is termed white noise if

$$c_{xx}[k] = \sigma^2 \delta[k]$$

where $\delta[k]$ is the discrete impulse function, $\delta[k] = \begin{cases} 1 & \text{if } k=0 \\ 0 & \text{o/w.} \end{cases}$

- The PSD $S_x(f)$ for white noise is given by $S_x(f) = \sigma^2 + M^2 \sum \delta(f)$

$$S_x(f) = \sum_{k=-\infty}^{\infty} c_{xx}[k] e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} (c_{xx}[k] + M^2) e^{j2\pi f k} = \sum_{k=-\infty}^{\infty} \sigma^2 \delta[k] e^{j2\pi f k} + \sum_{k=-\infty}^{\infty} M^2 e^{j2\pi f k} = \sigma^2 + M^2 \sum \delta(f)$$

for zero mean white noise, $M=0$, so $S_x(f) = \sigma^2$, i.e. flat across all freq. ← which for ARMA is zero mean white noise

- If the values x_n are drawn independently from a Gaussian distribution $N(0, \sigma^2)$, we have white Gaussian noise (WGN), the n th order pdf for WGN process is

$$f_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n N(x_i; 0, \sigma^2) \quad \leftarrow \text{simple product as } x_i \text{ are independent}$$

where $N(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ is the univariate normal pdf.

- The WGN process is SSS, since

$$f_{x_1, x_2, \dots, x_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n N(x_i; 0, \sigma^2) = f_{x_1} f_{x_2} \dots f_{x_n}(x_1, x_2, \dots, x_n)$$

Optimal filtering and signal detection

Optimal filtering

PSD and transfer function in terms of S_x .

- The PSD and TF can be expressed in terms of the normalised freq. Ω , (in radians per sample).

where $\Omega = \omega T$, and T is the sampling interval of the discrete-time process.

$$S_x(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} r_{xx}[k] e^{-j\Omega k}$$

$$r_{xx}[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_x(e^{j\Omega}) e^{j\Omega k} d\Omega$$

$$S_{xy}(e^{j\Omega}) = \sum_{k=-\infty}^{\infty} r_{xy}[k] e^{-j\Omega k}$$

$$r_{xy}[k] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{xy}(e^{j\Omega}) e^{j\Omega k} d\Omega$$

$$P_{[x][y]} = r_{xx}[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_x(e^{j\Omega}) d\Omega$$

$$P_{[w][z]} = \frac{1}{\pi} \int_{-\pi}^{\pi} S_x(e^{j\Omega}) d\Omega$$

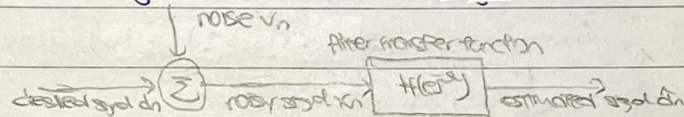
+ In this section, we explicitly write the PSD/TF as a function of $e^{j\Omega}$ to stress that it is always periodic w/ a period of 2π . $\rightarrow$ looks like a z-transform, w/ $z = e^{j\Omega}$

Optimal filtering.

- Optimal filtering is where we design filters that are optimally adapted to the statistical characteristics of a random process.

- The Wiener filter can optimally estimate a desired signal d_n given the noisy observations x_n , w/ some assumptions about the statistics of the random signal and noise processes.

- The general Wiener filtering problem can be expressed as a block diagram.



i.e. we have $x_n = d_n + v_n$ and $\hat{d}_n = x_n + h_n$.

- We define the error signal z_n as

$$z_n = d_n - \hat{d}_n = d_n - x_n + h_n$$

and the mean-squared error (MSE) J as

$$J = E[z_n^2]$$

where the expectation is wrt the random signal d_n and random noise v_n .

- The Wiener filter minimises the MSE J wrt the filter coefficients $\{h_n\}$.

- The Wiener filter assumes the following.

(i) $\{x_n\}$ and $\{d_n\}$ are jointly WSS, i.e. $r_{xx}[n+m] \rightarrow r_{xx}[m-n]$; $r_{dd}[n+m] \rightarrow r_{dd}[m-n]$, $r_{xd}[n+m] \rightarrow r_{xd}[m-n]$

(ii) $\{d_n\}$ and $\{v_n\}$ have zero mean, i.e. $E[d_n] = 0$, $E[v_n] = 0$

+ The Wiener filter is only the optimal linear estimator for stationary signals

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The general Wiener filter.

- In the general case, we can filter the observed signal x_n w/ an infinite-dimensional filter, having a non-causal impulse response $\{h_p\}$. Our estimate of the desired signal $\hat{d}_n$ is therefore

$$\hat{d}_n = \sum_{p=-\infty}^{\infty} h_p x_{n-p}$$

- We want to minimise the MSE J wrt the impulse response values h_p .

A sufficient condition for a min. is that simultaneously $\forall q \in (-\infty, \infty)$,

$$\frac{\partial J}{\partial h_q} = \frac{\partial E[\hat{e}_n^2]}{\partial h_q} = E\left[\frac{\partial \hat{e}_n^2}{\partial h_q}\right] = E[2\hat{e}_n \frac{\partial \hat{e}_n}{\partial h_q}] = 0.$$

Noting that $\frac{\partial \hat{e}_n}{\partial h_q} = \frac{\partial}{\partial h_q} [d_n - \sum_{p=-\infty}^{\infty} h_p x_{n-p}] = -x_{n-q}$, we get the condition

$$E[2\hat{e}_n \frac{\partial \hat{e}_n}{\partial h_q}] = E[2\hat{e}_n x_{n-q}] = 0 \rightarrow E[\hat{e}_n x_{n-q}] = 0, \forall q \in (-\infty, \infty)$$

orthogonality principle

Substituting $\hat{e}_n$ in $E[\hat{e}_n x_{n-q}]$ gives

$$E[\hat{e}_n x_{n-q}] = E[(d_n - \sum_{p=-\infty}^{\infty} h_p x_{n-p}) x_{n-q}] = E[d_n x_{n-q}] - \sum_{p=-\infty}^{\infty} h_p E[x_{n-p} x_{n-q}] = r_{dx}[q] - \sum_{p=-\infty}^{\infty} h_p r_{xx}[q-p]$$

$$r_{dx}[q] - \sum_{p=-\infty}^{\infty} h_p r_{xx}[q-p] = 0 \rightarrow \sum_{p=-\infty}^{\infty} h_p r_{xx}[q-p] = r_{dx}[q], \forall q \in (-\infty, \infty)$$

which are collectively known as the Wiener-Hopf eqns.

- The simplest way to solve for the infinite no. of unknowns h_p in the Wiener-Hopf eqns is to transform everything into the frequency domain.

Rewriting the Wiener-Hopf eqns as a convolution,

$$h_q * r_{xx}[q] = r_{dx}[q] \quad \forall q \in (-\infty, \infty)$$

Taking the DTFT on both sides,

$$H(e^{j\omega}) S_x(e^{j\omega}) = S_{dx}(e^{j\omega}) \rightarrow H(e^{j\omega}) = \frac{S_{dx}(e^{j\omega})}{S_x(e^{j\omega})}$$

- The MSE of the optimal filter, J_{min} , can be found using (i) time or (ii) frequency domain

$$(i). J = E[\hat{e}_n^2] = E[\hat{e}_n (d_n - \sum_{p=-\infty}^{\infty} h_p x_{n-p})] = E[\hat{e}_n d_n] - \sum_{p=-\infty}^{\infty} h_p E[\hat{e}_n x_{n-p}]$$

For the optimal filter, the orthogonality principle states $E[\hat{e}_n x_{n-p}] = 0 \forall p \in (-\infty, \infty)$, so

$$J_{min} = E[\hat{e}_n d_n] = E[(d_n - \sum_{p=-\infty}^{\infty} h_p x_{n-p}) d_n] = E[d_n^2] - \sum_{p=-\infty}^{\infty} h_p E[d_n x_{n-p}] = r_{dd}[0] - \sum_{p=-\infty}^{\infty} h_p r_{dx}[p]$$

$$(ii) \hat{e}_n \hat{e}_{n+k} = (d_n - \sum_{p=-\infty}^{\infty} h_i x_{n-i}) (d_{n+k} - \sum_{j=-\infty}^{\infty} h_j x_{n+k-j}) = d_n d_{n+k} - d_n \sum_{j=-\infty}^{\infty} h_j x_{n+k-j} - d_{n+k} \sum_{i=-\infty}^{\infty} h_i x_{n-i} + \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_i h_j x_{n-i} x_{n+k-j}$$

$$r_{\hat{e}\hat{e}}[k] = E[\hat{e}_n \hat{e}_{n+k}] = r_{dd}[k] - \sum_{j=-\infty}^{\infty} h_j r_{dx}[k-j] - \sum_{i=-\infty}^{\infty} h_i r_{dx}[i-k] + \sum_{i=-\infty}^{\infty} \sum_{j=-\infty}^{\infty} h_i h_j r_{xx}[k+i-j]$$

$$= r_{dd}[k] - h_k * r_{dx}[k] - \hat{h}_k * r_{dx}[k] + h_k * \hat{h}_k * r_{xx}[k]$$

$$S_{\hat{e}\hat{e}}(e^{j\omega}) = S_d(e^{j\omega}) - H(e^{j\omega}) S_{dx}(e^{j\omega}) - H^*(e^{j\omega}) S_{dx}(e^{j\omega}) + H(e^{j\omega}) H^*(e^{j\omega}) S_x(e^{j\omega})$$

$$J = E[\hat{e}_n^2] = r_{\hat{e}\hat{e}}[0] = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{\hat{e}\hat{e}}(e^{j\omega}) d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} [S_d(e^{j\omega}) - H(e^{j\omega}) S_{dx}(e^{j\omega}) - H^*(e^{j\omega}) S_{dx}(e^{j\omega}) + |H(e^{j\omega})|^2 S_x(e^{j\omega})] d\omega$$

For the optimal filter, solving the Wiener-Hopf eqns gives $H(e^{j\omega}) = \frac{S_{dx}(e^{j\omega})}{S_x(e^{j\omega})}$

$$H^*(e^{j\omega}) S_{dx}(e^{j\omega}) = \frac{S_{dx}^*(e^{j\omega}) S_{dx}(e^{j\omega})}{S_x^*(e^{j\omega})} = \frac{|S_{dx}(e^{j\omega})|^2}{S_x^*(e^{j\omega})} ; |H(e^{j\omega})|^2 S_x(e^{j\omega}) = \frac{|S_{dx}(e^{j\omega})|^2}{S_x(e^{j\omega})}$$

$$\rightarrow J_{min} = \frac{1}{2\pi} \int_{-\pi}^{\pi} [S_d(e^{j\omega}) - |H(e^{j\omega})|^2 S_x(e^{j\omega})] d\omega$$

+ If the random process is ergodic, we estimate $r_{xx}[k]$, $r_{dx}[k]$ from time averages

• In general, the optimal filter yielded is non-causal $\rightarrow$ not implementable in practice

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The FIR Wiener filter

- For the FIR Wiener filter, it has a causal p -th order finite impulse response $\{h_p\}_{p=0}^P$. Our estimate of the desired signal, $\hat{d}_n$, is therefore

$$\hat{d}_n = \sum_{p=0}^P h_p x_{n-p}$$

- As before, we want to minimize the MSE J wrt the impulse response values h_q .

orthogonality principle: $E[\hat{e}_n x_{n-q}] = 0, \forall q \in [0, P]$

Wiener-Hopf eqns: $\sum_{p=0}^P h_p r_{xx}[q-p] = r_{xd}[q], \forall q \in [0, P]$

- The Wiener-Hopf eqns are now a finite set of simultaneous eqns that can be solved in the time domain. The eqns may be written in matrix form as

$$\underline{R}_x \underline{h} = \underline{r}_{xd}$$

where $\underline{h} = \begin{bmatrix} h_0 \\ h_1 \\ \vdots \\ h_P \end{bmatrix}$, $\underline{r}_{xd} = \begin{bmatrix} r_{xd}[0] \\ r_{xd}[1] \\ \vdots \\ r_{xd}[P] \end{bmatrix}$, $\underline{R}_x = \begin{bmatrix} r_{xx}[0] & r_{xx}[1] & \dots & r_{xx}[P] \\ r_{xx}[1] & r_{xx}[0] & \dots & r_{xx}[P-1] \\ \vdots & \vdots & \ddots & \vdots \\ r_{xx}[P] & r_{xx}[P-1] & \dots & r_{xx}[0] \end{bmatrix}$

(Note the correlation matrix $\underline{R}_x$ is symmetric and has const. diagonals).

the coefficient vector can be found by matrix inversion,

$$\underline{h} = \underline{R}_x^{-1} \underline{r}_{xd}$$

- As before, the MSE of the optimal filter J_{min} , is given by

$$J_{min} = r_{dd}[0] - \sum_{p=0}^P h_p r_{xd}[p] = r_{dd}[0] - \underline{r}_{xd}^T \underline{h} = r_{dd}[0] - \underline{r}_{xd}^T \underline{R}_x^{-1} \underline{r}_{xd}$$

Uncorrelated signal and noise processes,

- Typically, the environmental noise v_n is independent of the desired signal $d_n \rightarrow$ uncorrelated, i.e.

$$r_{dv}[k] = E[d_n v_{n+k}] = 0, \quad k \in (-\infty, \infty)$$

- In this situation, the correlation functions $r_{xx}[k]$ and $r_{xd}[k]$ can be simplified.

(i) $r_{xd}[k] = E[d_n x_{n+k}] = E[(d_n + v_n) x_{n+k}] = E[d_n x_{n+k}] + E[v_n x_{n+k}] = r_{dd}[k]$

$$r_{xd}[k] = r_{dd}[k] \quad \leftrightarrow \quad S_{xd}(e^{j\omega}) = S_{dd}(e^{j\omega})$$

(ii) $r_{xx}[k] = E[x_n x_{n+k}] = E[(d_n + v_n)(d_{n+k} + v_{n+k})] = E[d_n d_{n+k}] + E[d_n v_{n+k}] + E[v_n d_{n+k}] + E[v_n v_{n+k}] = r_{dd}[k] + r_{vv}[k]$

$$r_{xx}[k] = r_{dd}[k] + r_{vv}[k] \quad \leftrightarrow \quad S_x(e^{j\omega}) = S_d(e^{j\omega}) + S_v(e^{j\omega})$$

- For the general Wiener filter, the optimal filter TF $H(e^{j\omega})$ becomes,

$$H(e^{j\omega}) = \frac{S_{xd}(e^{j\omega})}{S_x(e^{j\omega})} = \frac{S_{dd}(e^{j\omega})}{S_d(e^{j\omega}) + S_v(e^{j\omega})} = \frac{1}{1 + 1/p(\omega)}$$

where $p(\omega) = \frac{S_v(e^{j\omega})}{S_d(e^{j\omega})}$ is the (freq.-dependent) SNR power ratio.

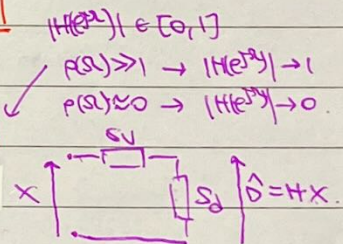
- The MSE of the optimal filter, J_{min} is now reduced to,

$$J_{min} = \frac{1}{2\pi} \int_{-\pi}^{\pi} [S_{dd}(e^{j\omega}) + H(e^{j\omega}) S_v(e^{j\omega})] d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{dd}(e^{j\omega}) \left(1 - \frac{1}{1 + 1/p(\omega)}\right) d\omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} S_{dd}(e^{j\omega}) \frac{1}{1 + p(\omega)} d\omega$$

- In practice, the correlation functions are estimated as (assuming signal is ergodic).

$$\hat{r}_{xx}[k] \approx \frac{1}{N} \sum_{n=0}^{N-k} x_n x_{n+k}, \quad \hat{r}_{dd}[k] = \hat{r}_{xx}[k] - \hat{r}_{vv}[k] = \begin{cases} \hat{r}_{xx}[k] - \sigma_v^2 & \text{if } k=0 \\ \hat{r}_{xx}[k] & \text{o.w.} \end{cases}$$

* We should make sure $\hat{r}_{xx}[k]$ and $\hat{r}_{dd}[k]$ are valid autocorrelation sequences, since they are only empirical estimates.



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Validity of autocorrelation sequences

- For the autocorrelation $r_{xx}[k]$ to form a valid sequence for constructing the correlation matrix R_x , the necessary condition is the resulting R_x is positive-semidefinite, i.e.

$$\underline{a}^T R_x \underline{a} \geq 0$$

for any length $N+1$ vector $\underline{a}$.

(since R_x is symmetric, this means its eigenvalues λ_i must satisfy $\lambda_i \geq 0 \forall i$)

- consider the vector $\underline{x}_n = [x_n \ x_{n-1} \ \dots \ x_{n-P}]^T$ for real-valued x_n 's, $\underline{a}^T \underline{x}_n = \underline{x}_n^T \underline{a}$

$$0 \leq (\underline{a}^T \underline{x}_n)^2 = (\underline{a}^T \underline{x}_n)(\underline{x}_n^T \underline{a}) = \underline{a}^T (\underline{x}_n \underline{x}_n^T) \underline{a}$$

The (i,j) th element of $\underline{x}_n \underline{x}_n^T$ is $x_{n-i+1} x_{n-j+1}$ and $E[x_{n-i+1} x_{n-j+1}] = r_{xx}[i-j] \rightarrow E[\underline{x}_n \underline{x}_n^T] = R_x$.

$\therefore 0 \leq E[\underline{a}^T (\underline{x}_n \underline{x}_n^T) \underline{a}] = \underline{a}^T E[\underline{x}_n \underline{x}_n^T] \underline{a} = \underline{a}^T R_x \underline{a} \rightarrow R_x$ must be positive-semidefinite.

extending the Wiener filter

- To extend the Wiener filter beyond the regular noise reduction case, we simply replace the desired signal d w/ what we want to predict/estimate, then rederive the filter.
- In general, the FIR formula

$$\underline{h} = R_x^{-1} r_{xd}$$

applies for any form of the (unobserved) desired signal d and the (observed) "noisy" signal x , provided we can calculate the necessary correlation functions.

- Some standard examples include:

↳ Prediction of a signal, $x_n = u_n + v_n$, $d_n = u_{n+p}$

↳ Smoothing of a signal, $x_n = u_n + v_n$, $d_n = u_{n-p}$

↳ Deconvolution, $x_n = h_{g,n} * u_n + v_n = \sum_{q=0}^Q h_{g,q} u_{n-q} + v_n$, $d_n = u_n$.

- In all cases, the Wiener filter estimate takes the same form as before

$$\hat{d}_n = \sum_{p=0}^P h_p x_{n-p}$$

and the error is defined as before

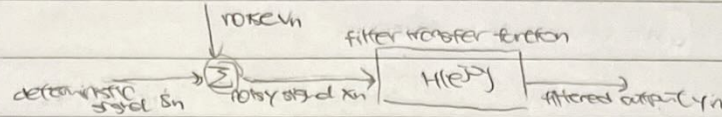
$$\epsilon_n = d_n - \hat{d}_n$$

Note that only d is different, and we need to rederive an expression for $r_{xx}[k]/S_{xx}(e^{j\omega})$.

signal detection

Signal detection

- The matched filter is an optimal FIR filter that can detect a deterministic signal s_n , $n=0, \dots, N-1$, from the noisy observations x_n .
- The general signal detection problem can be expressed as a block diagram



i.e. we have $x_n = s_n + v_n$ and $y_n = x_n + h_n$

- In vector form, we have

$$\underline{x} = \underline{s} + \underline{v}$$

where $\underline{s} = [s_0, s_1, \dots, s_{N-1}]^T$ and $\underline{x} = [x_0, x_1, \dots, x_{N-1}]^T$.

- Using a $(N-1)$ th order FIR filter, the o/p of the filter at time $N-1$ is given by

$$y_{N-1} = \sum_{m=0}^{N-1} h_m x_{N-1-m} = \underline{h}^T \underline{x} = \underline{h}^T (\underline{s} + \underline{v}) = \underline{h}^T \underline{s} + \underline{h}^T \underline{v} = \sum_{m=0}^{N-1} h_m s_{N-1-m} + \sum_{m=0}^{N-1} h_m v_{N-1-m} = y_{N-1}^s + y_{N-1}^v$$

where y_{N-1}^s is the o/p from the signal only part; y_{N-1}^v is the o/p from the noise only part.

- We define the o/p SNR as

$$\text{SNR} = \frac{E[y_{N-1}^s]^2]}{E[y_{N-1}^v]^2]} = \frac{E[\underline{h}^T \underline{s}]^2]}{E[\underline{h}^T \underline{v}]^2]} = \frac{|\underline{h}^T \underline{s}|^2]}{E[\underline{h}^T \underline{v}]^2]}$$

- The matched filter maximises the o/p SNR wrt the filter coefficients $\{h_p\}$ to give the best possible chance of detecting the signal s_n .

Signal output energy.

- The signal o/p energy is given by $y_{N-1}^{s^2} = |\underline{h}^T \underline{s}|^2 = (\underline{h}^T \underline{s})(\underline{s}^T \underline{h}) = \underline{h}^T (\underline{s} \underline{s}^T) \underline{h}$.
- Consider the matrix $\underline{M} = \underline{s} \underline{s}^T$. We want to find its eigenvectors $\underline{u}$ and eigenvalues λ .

$$\underline{u} = \underline{s} : \quad \underline{M} \underline{s} = (\underline{s} \underline{s}^T) \underline{s} = \underline{s} (\underline{s}^T \underline{s}) = (\underline{s}^T \underline{s}) \underline{s}$$

i.e. $\underline{u}_0 = \frac{\underline{s}}{\|\underline{s}\|}$ is an eigenvector and $\lambda_0 = \underline{s}^T \underline{s}$ is the corresponding eigenvalue.

$$\underline{u} = \{\underline{u}' : \underline{s}^T \underline{u}' = 0\} : \quad \underline{M} \underline{u}' = (\underline{s} \underline{s}^T) \underline{u}' = \underline{s} (\underline{s}^T \underline{u}') = 0$$

i.e. $\underline{u} = \underline{u}'$ is an eigenvector, w/ $\lambda = 0$. (There are $N-1$ orthogonal vectors orthogonal to $\underline{s}$, $\underline{u}_1, \underline{u}_2, \dots, \underline{u}_{N-1}$)

- We can represent any filter coefficient vector $\underline{h}$ as a linear combination of these eigenvectors.

$$\underline{h} = \alpha \underline{u}_0 + \beta \underline{u}_1 + \gamma \underline{u}_2 + \dots + \dots \underline{u}_{N-1}$$

$$\rightarrow \underline{M} \underline{h} = \underline{M} (\alpha \underline{u}_0 + \beta \underline{u}_1 + \gamma \underline{u}_2 + \dots + \dots \underline{u}_{N-1}) = \alpha \underline{M} \underline{u}_0 + \beta \underline{M} \underline{u}_1 + \gamma \underline{M} \underline{u}_2 + \dots + \dots \underline{M} \underline{u}_{N-1} = \alpha (\underline{s} \underline{s}^T) \underline{u}_0$$

- The signal o/p energy is therefore

$$\underline{h}^T \underline{M} \underline{h} = \underline{h}^T \underline{M} \underline{h} = \underline{h}^T (\alpha \underline{s} \underline{s}^T \underline{u}_0) = (\alpha \underline{s}^T \underline{s}) \underline{h}^T \underline{u}_0 = (\alpha \underline{s}^T \underline{s}) (\alpha \underline{u}_0 + \beta \underline{u}_1 + \gamma \underline{u}_2 + \dots + \dots \underline{u}_{N-1})^T \underline{u}_0$$

$$\therefore y_{N-1}^{s^2} = |\underline{h}^T \underline{s}|^2 = \alpha^2 \underline{s}^T \underline{s}$$

since $\underline{u}_0^T \underline{u}_0 = 1$ and $\underline{u}_0^T \underline{u}' = 0$ (orthogonality).

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Noise output energy

- The expected noise o/p energy is given by $E[y_{N+1}^2] = E[h^T \mathcal{V} \mathcal{V}^T h] = E[h^T \mathcal{V} \mathcal{V}^T] h = h^T E[\mathcal{V} \mathcal{V}^T] h$

- considering the case where the noise is white, for time indices $i=0,1,\dots,N-1$ and $j=0,1,\dots,N-1$,

$$E[v_i v_j^T] = \begin{cases} \sigma_v^2 & \text{if } i=j \\ 0 & \text{o/w} \end{cases}$$

$\therefore E[\mathcal{V} \mathcal{V}^T] = \sigma_v^2 \mathbf{I}$, where $\mathbf{I}$ is the $N \times N$ identity matrix.

- The expected noise o/p energy is therefore

$$E[y_{N+1}^2] = h^T E[\mathcal{V} \mathcal{V}^T] h = \sigma_v^2 h^T h$$

expanding h in terms of the eigenvector of $\mathbf{I} = \mathbf{I}^T$, ($h = \alpha v_0 + \beta v_1 + \gamma v_2 + \dots + v_{N-1}$), we have

$$E[y_{N+1}^2] = \sigma_v^2 h^T h = \sigma_v^2 (\alpha^2 + \beta^2 + \gamma^2 + \dots)$$

SNR maximisation.

- The SNR may now be expressed as

$$\text{SNR} = \frac{|h^T \mathbf{I} h|^2}{E[y_{N+1}^2]} = \frac{\alpha^2 \mathbf{I}^T \mathbf{I} \mathbf{I}}{\sigma_v^2 (\alpha^2 + \beta^2 + \gamma^2 + \dots)}$$

- By inspection, scaling h by some factor p will not change the SNR, since both the numerator and denominator will both scale equally by p^2 . $\rightarrow$ we can fix $|h| = |\alpha v_0 + \beta v_1 + \gamma v_2 + \dots + v_{N-1}| = 1$, then max.

- setting $|h| = 1$, we have $(\alpha^2 + \beta^2 + \gamma^2 + \dots) = 1$, so the SNR becomes

$$\text{SNR} = \frac{\alpha^2 \mathbf{I}^T \mathbf{I} \mathbf{I}}{\sigma_v^2}$$

- The largest possible value of α given that $|h| = 1$ corresponds to $\alpha = 1$ (i.e. $\beta = \gamma = \dots = 0$).

This means the optimal filter coefficient vector h is

$$h = 1 v_0 = \frac{\mathbf{I} \mathbf{I}}{|\mathbf{I}|}$$

time-reversed signal, normalised.

and the corresponding SNR is

$$\text{SNR} = \frac{\mathbf{I}^T \mathbf{I} \mathbf{I}}{\sigma_v^2}$$

depends on energy of signal s and noise v , as expected.

Practical implementation of the matched filter.

- For a signal s with length N , to optimise a filter h , we considered a batch of data x of length N .

- In practice, we would run this optimised filter over a much larger length of data $\{x_k\}$, which contains s at some unknown position, then find the time at which max. energy occurs.

- The time at which max. energy occurs is the pt at which s can be detected, based on some threshold.

- Matched filtering increases the best SNR achievable significantly. Before filtering,

$$\text{SNR}_{\text{max}} = \frac{\text{max signal value}^2}{\text{average noise energy}} = \frac{\text{max } s_n}{\sigma_v^2}$$

Estimation theory and inference For Personal Use Only -bkwk2

Estimation theory and inference methods.

Estimation and inference

- In estimation theory, we start w/ a vector of signal measurements $\underline{x} = [x_0 \ x_1 \ \dots \ x_{N-1}]^T$ and some unknown quantities (parameters) that we wish to infer, $\underline{\theta} = [\theta_0 \ \theta_1 \ \dots \ \theta_{p-1}]^T$ (usually $p \ll N$)
- We typically suppose the probability distribution of the data $\underline{x}$ can be expressed as a likelihood function $p(\underline{x}|\underline{\theta})$, which represents how "likely" diff. realisations of the observed data would be if we knew $\underline{\theta}$.
- In many problems, we may treat $\underline{\theta}$ as a random vector, and a prior pdf can be formulated for $\underline{\theta}$, $p(\underline{\theta})$. the prior represents prior belief about likely parameter configurations prior to any data observations.
- The prior distribution can be used to regularise the inference problem by constraining the parameter search to reasonable parts of the domain of $\underline{\theta}$ — Bayesian inference.
- In some cases, we may not wish to consider $\underline{\theta}$ as a random quantity. $\rightarrow$ we can only rely on the likelihood function — classical inference.
- Estimation is the task of formulating an estimate $\hat{\underline{\theta}}$ which is in some sense close to the true $\underline{\theta}$. Inference has estimation as a special case, and studies the whole prob. distribution of the unknown.

Simple estimators and their properties

- It is useful to consider some fundamental properties of estimators — their mean and variance.
- An estimator $\hat{\theta}$ is said to be unbiased if its expectation satisfies

$$E[\hat{\theta}] = \theta$$

- An estimator $\hat{\theta}$ is said to be consistent if it is unbiased and its variance satisfies

$$\lim_{N \rightarrow \infty} \text{Var}[\hat{\theta}] = 0$$

- The defn of unbiased estimators and their variance can be used to measure the performance of any proposed estimation scheme.
- Estimators can be designed for a given problem specifically to lead to no bias and min variance.
- consider a set of N independent samples of a RV X which has mean μ and standard deviation σ

$$\{x_1, x_2, x_3, \dots, x_N\}$$

An estimator for the mean $\hat{\mu}$, is obtained by taking the arithmetic average of the samples.

$$\hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i$$

- The expectation of the estimator $\hat{\mu}$, $E[\hat{\mu}]$ is given by

$$E[\hat{\mu}] = E\left[\frac{1}{N} \sum_{i=1}^N x_i\right] = \frac{1}{N} \sum_{i=1}^N E[x_i] = \mu \rightarrow \text{unbiased}.$$

The variance of the estimator $\hat{\mu}$, $\text{Var}[\hat{\mu}]$ is given by

$$\text{Var}[\hat{\mu}] = E[(\hat{\mu} - E[\hat{\mu}])^2] = E[\hat{\mu}^2] - E[\hat{\mu}]^2 = (\mu^2 + \frac{\sigma^2}{N}) - \mu^2 = \frac{\sigma^2}{N} \rightarrow \text{tends to 0 as } N \rightarrow \infty.$$

where $E[\hat{\mu}^2] = E\left[\left(\frac{1}{N} \sum_{i=1}^N x_i\right)\left(\frac{1}{N} \sum_{j=1}^N x_j\right)\right] = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N E[x_i x_j]$, and $E[x_i x_j] = \begin{cases} E[x_i^2] = \mu^2 + \sigma^2 & i=j \\ E[x_i]E[x_j] = \mu^2 & i \neq j \end{cases}$.

$$\therefore E[\hat{\mu}^2] = \frac{1}{N^2} [(N^2 - N)\mu^2 + N(\mu^2 + \sigma^2)] = \frac{1}{N^2} (N^2\mu^2 + N\sigma^2) = \mu^2 + \frac{\sigma^2}{N}.$$

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The general linear model / linear regression model.

- In the linear model, it is assumed that the data x are generated as a linear function of the parameters θ w/ an additive random modelling error term e_n .

$$x_n = g_n^T \theta + e_n$$

where g_n is a P -dimensional column vector.

- This can be written for the whole vector x as

$$x = G\theta + e, \quad \text{where } G = \begin{bmatrix} g_1^T \\ g_2^T \\ \vdots \\ g_N^T \end{bmatrix}$$

- The choice of the design matrix G will lead to a wide range of possible models, for example,

① constant level in noise

$$x_n = \theta + e_n \quad \leftrightarrow \quad x = G\theta + e, \quad \text{where } G = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

② sinusoidal mode (1 frequency)

$$x_n = a \cos(n\omega) + b \sin(n\omega) + e_n \quad \leftrightarrow \quad x = G\theta + e, \quad \text{where } G = [\underline{c}(\omega) \quad \underline{s}(\omega)], \quad \theta = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$(\text{note } \underline{c}(\omega) = [\cos(0) \quad \cos(\omega) \quad \dots \quad \cos((N-1)\omega)]^T, \quad \underline{s}(\omega) = [\sin(0) \quad \sin(\omega) \quad \dots \quad \sin((N-1)\omega)]^T)$$

③ sinusoidal mode (J frequencies)

$$x_n = \sum_{j=1}^J a_j \cos(n\omega_j) + b_j \sin(n\omega_j) + e_n \quad \leftrightarrow \quad x = G\theta + e, \quad \text{where } G = [\underline{c}(\omega_1) \quad \underline{s}(\omega_1) \quad \dots \quad \underline{c}(\omega_J) \quad \underline{s}(\omega_J)], \quad \theta = \begin{bmatrix} a_1 \\ b_1 \\ \vdots \\ a_J \\ b_J \end{bmatrix}$$

$$(\text{note } \underline{c}(\omega_j) = [\cos(0) \quad \cos(\omega_j) \quad \dots \quad \cos((N-1)\omega_j)]^T, \quad \underline{s}(\omega_j) = [\sin(0) \quad \sin(\omega_j) \quad \dots \quad \sin((N-1)\omega_j)]^T)$$

④ Autoregressive model.

$$x_n = \sum_{i=1}^p a_i x_{n-i} + e_n \quad \leftrightarrow \quad x = Gg + e, \quad \text{where } G = \begin{bmatrix} x_0 & x_{-1} & \dots & x_{-(p-1)} & x_{-p} \\ x_1 & x_0 & \dots & x_{-(p-2)} & x_{-(p-1)} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{N-1} & x_{N-2} & \dots & x_{N-p} & x_{N-p-1} \end{bmatrix}$$

(Note G is not fixed before making the measurements, but is made up of observed data pts.)

$\rightarrow$ cannot directly generate a vector of data using the formula $x = Gg + e \rightarrow$ generated sequentially)

Einstein-Wigner-Khinchin theorem.

- the Einstein-Wigner-Khinchin thm. states that the power spectrum is equal to the expected value of the time-normalised DTFT-squared of the signal values, i.e.

$$S_x(e^{j\omega}) = \lim_{N \rightarrow \infty} E \left[\frac{1}{2N+1} |X^N(e^{j\omega})|^2 \right]$$

- consider a time-windowed version of a signal x_n , w/ duration of $2N+1$ samples, denoted by x_n^N ,

$$x_n^N = w_n^N x_n \quad \text{where } w_n^N = \begin{cases} 1 & \text{if } -N \leq n \leq N \\ 0 & \text{otherwise} \end{cases}$$

Its DTFT $X^N(e^{j\omega})$ is just the DFT of a frame of $2N+1$ data pts + linear phase shift

$$\text{DTFT}[x_n^N] = X^N(e^{j\omega}) = \sum_{n=-N}^N w_n^N x_n e^{j\omega n}$$

$$\therefore |X^N(e^{j\omega})|^2 = X^N(e^{j\omega}) X^N(e^{j\omega})^* = \text{DTFT}[x_n^N + \hat{x}_n^N] = \text{DTFT} \left[\sum_{m=-N}^N x_m w_m^N x_{m-n} w_{m-n}^N \right]$$

ensures we have finite energy
Dividing both sides by the window duration $2N+1$ and taking expectations,

$$E \left[\frac{1}{2N+1} |X^N(e^{j\omega})|^2 \right] = E \left[\frac{1}{2N+1} \text{DTFT} \left[\sum_{m=-N}^N x_m w_m^N x_{m-n} w_{m-n}^N \right] \right] = \frac{1}{2N+1} \text{DTFT} \left[\sum_{m=-N}^N E[x_m w_m^N x_{m-n} w_{m-n}^N] \right] = \text{DTFT}[c(n)]$$

where $c(n) = \frac{1}{2N+1} \sum_{m=-N}^N w_m^N w_{m-n}^N$ is the deterministic autocorrelation function of the window function w_n .

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Noting that $\text{DFT}[r_x[n]] = S_x(e^{j\omega})$ and defining $\text{DFT}[t[n]] = T(e^{j\omega})$, we have

$$E\left[\frac{1}{N} |X(e^{j\omega})|^2\right] = \text{DFT}[r_x[n]t[n]] = S_x(e^{j\omega}) * T(e^{j\omega})$$

As N increases, $t[n]$ become flatter and $T(e^{j\omega})$ tends to a delta function, so the limit as $N \rightarrow \infty$ is

$$\lim_{N \rightarrow \infty} E\left[\frac{1}{N} |X(e^{j\omega})|^2\right] = \lim_{N \rightarrow \infty} \text{DFT}[r_x[n]t[n]] = \text{DFT}[r_x[n]] = S_x(e^{j\omega})$$

i.e. power spectrum is prop. to expected value of the DFT squared of the data.

Useful linear algebra/vector calculus properties

- Denote $\nabla J = \frac{\partial J}{\partial \underline{x}}$ as the vector gradient for a column vector $\underline{x}$, we have

① $J = \underline{b}^T \underline{x} \rightarrow \nabla J = \underline{b}$ for a column vector $\underline{b}$.

$$J = \underline{b}^T \underline{x} = \sum_i b_i x_i \rightarrow \frac{\partial J}{\partial x_i} = b_i \rightarrow \frac{\partial J}{\partial \underline{x}} = \underline{b}$$

② $J = \underline{x}^T \underline{B} \underline{x} \rightarrow \nabla J = (\underline{B} + \underline{B}^T) \underline{x}$ for a matrix $\underline{B}$.

$$J = \underline{x}^T \underline{B} \underline{x} = \sum_i x_i \left(\sum_j B_{ij} x_j \right) \rightarrow \frac{\partial J}{\partial x_i} = 2B_{ii} x_i + \sum_{j \neq i} (x_j B_{ji} + B_{ij} x_i) = \sum_j (B_{ij} + B_{ji}) x_j$$

$$\therefore \frac{\partial J}{\partial \underline{x}} = (\underline{B} + \underline{B}^T) \underline{x}$$

* for symmetric $\underline{B}$, $\underline{B}^T = \underline{B}$, so $\frac{\partial J}{\partial \underline{x}} = (\underline{B} + \underline{B}) \underline{x} = 2\underline{B} \underline{x}$

- For a matrix in the form $\underline{B} = \underline{G}^T \underline{G}$, it is positive semi-definite, i.e. $\underline{x}^T \underline{B} \underline{x} \geq 0$ for any $\underline{x}$.

$$0 \leq (\underline{G} \underline{x})^T (\underline{G} \underline{x}) = \underline{x}^T \underline{G}^T \underline{G} \underline{x} = \underline{x}^T \underline{B} \underline{x}$$

If $\underline{G}$ is full rank, then $\underline{G} \underline{x} \neq 0$ for any nonzero $\underline{x}$, so $\underline{B}$ is positive definite, i.e. $\underline{x}^T \underline{B} \underline{x} > 0$ for any $\underline{x} \neq 0$

- The global min. of $J = \underline{x}^T \underline{B} \underline{x} - 2\underline{b}^T \underline{x}$ wrt. $\underline{x}$ for a symmetric positive definite $\underline{B}$ is $\underline{x} = \underline{B}^{-1} \underline{b}$

(i) differentiating J using vector calculus.

$$\frac{\partial J}{\partial \underline{x}} = 2\underline{B} \underline{x} - 2\underline{b} = 0 \rightarrow \underline{x} = \underline{B}^{-1} \underline{b}$$

(ii) completing the square

$$J = \underline{x}^T \underline{B} \underline{x} - 2\underline{b}^T \underline{x} = (\underline{x} - \underline{B}^{-1} \underline{b})^T \underline{B} (\underline{x} - \underline{B}^{-1} \underline{b}) - \underline{b}^T \underline{B}^{-1} \underline{b} \rightarrow \underline{x} = \underline{B}^{-1} \underline{b}$$

Linear estimation of the general linear model

Least squares (LS) estimator $\underline{\theta}^{OLS}$

- For the general linear model $\underline{x} = \underline{G} \underline{\theta} + \underline{e}$, linear estimator finds the best fit by min. the error J ,

$$J = \sum_{n=0}^{N-1} e_n^2 = \underline{e}^T \underline{e}$$

expanding using $\underline{e} = \underline{x} - \underline{G} \underline{\theta}$,

$$J = \underline{e}^T \underline{e} = (\underline{x} - \underline{G} \underline{\theta})^T (\underline{x} - \underline{G} \underline{\theta}) = \underline{x}^T \underline{x} + \underline{\theta}^T \underline{G}^T \underline{G} \underline{\theta} - 2\underline{\theta}^T \underline{G}^T \underline{x}$$

The global min. of J wrt $\underline{\theta}$ can be found to be $\underline{\theta}^{OLS} = (\underline{G}^T \underline{G})^{-1} \underline{G}^T \underline{x}$

(i) $\frac{\partial J}{\partial \underline{\theta}} = 0 + 2\underline{G}^T \underline{G} \underline{\theta} - 2\underline{G}^T \underline{x} = 0 \rightarrow \underline{\theta}^{OLS} = (\underline{G}^T \underline{G})^{-1} \underline{G}^T \underline{x}$

(ii) $J = \underline{x}^T \underline{x} + \underline{\theta}^T \underline{G}^T \underline{G} \underline{\theta} - 2\underline{\theta}^T \underline{G}^T \underline{x} = \underline{\theta}^T (\underline{G}^T \underline{G}) \underline{\theta} - 2\underline{\theta}^T \underline{G}^T \underline{x} + \underline{x}^T \underline{x}$
 $\therefore \underline{\theta}^{OLS} = (\underline{G}^T \underline{G})^{-1} \underline{G}^T \underline{x}$

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Properties of the linear estimator

- The OLS estimator of the general linear model is a linear estimator since $\hat{\beta}$ is in the form $\hat{\beta} = Cx$.
- To investigate its properties, we consider its mean and variance.

① Mean, $E[\hat{\beta}^{OLS}]$

$$E[x] = E[G\theta + \epsilon] = E[G\theta] + E[\epsilon] = G\theta$$

$$\therefore E[\hat{\beta}^{OLS}] = E[(G^T G)^{-1} G^T x] = (G^T G)^{-1} G^T E[x] = (G^T G)^{-1} G^T G \theta = \theta \rightarrow \text{unbiased.}$$

② Covariance, $\text{cov}[\hat{\beta}^{OLS}] = E[(\hat{\beta}^{OLS} - E[\hat{\beta}^{OLS}])(\hat{\beta}^{OLS} - E[\hat{\beta}^{OLS}])^T]$

define the OLS matrix form as $C = (G^T G)^{-1} G^T$, and consider a general unbiased linear estimator

$$\hat{\beta} = D x \quad \text{where} \quad D = C + \Delta$$

for $\hat{\beta} = D x$ to be unbiased, we req.

$$E[\hat{\beta}] = E[D x] = \theta \rightarrow E[D x] = E[(C + \Delta)x] = (C + \Delta)E[x] = (C + \Delta)G\theta = \theta + \Delta G\theta = \theta$$

$$\therefore \Delta G\theta = 0, \text{ since } \theta \text{ must work for any } \theta.$$

θ is zero mean white noise w/ variance $\sigma_e^2 I$

consider the outer product $x x^T$, and take its expectation,

$$x x^T = (G\theta + \epsilon)(G\theta + \epsilon)^T = G\theta\theta^T G^T + \epsilon\epsilon^T + \epsilon\theta^T G^T + G\theta\epsilon^T \rightarrow E[x x^T] = G\theta\theta^T G^T + \sigma_e^2 I$$

for the linear estimator $\hat{\beta} = D x$, the expectation of the outer product $\hat{\beta}\hat{\beta}^T$, $E[\hat{\beta}\hat{\beta}^T]$ is

$$E[\hat{\beta}\hat{\beta}^T] = E[D x x^T D^T] = D E[x x^T] D^T = D (G\theta\theta^T G^T + \sigma_e^2 I) D^T = D G\theta\theta^T G^T D^T + \sigma_e^2 D D^T$$

$$\text{since } D G\theta = (C + \Delta)G\theta = C G\theta + \Delta G\theta = \theta \rightarrow D G\theta\theta^T G^T D^T = \theta\theta^T. \quad \Delta G\theta^T = 0 \quad - (G^T G)^{-1} G^T \Delta^T = 0$$

$$\therefore \text{cov}[\hat{\beta}] = E[\hat{\beta}\hat{\beta}^T] - \theta\theta^T = \theta\theta^T + \sigma_e^2 D D^T - \theta\theta^T = \sigma_e^2 (C + \Delta)(C + \Delta)^T = \sigma_e^2 (C C^T + \Delta \Delta^T + \Delta C^T + C \Delta^T)$$

$$= \sigma_e^2 (C C^T + \Delta \Delta^T) = \sigma_e^2 ((G^T G)^{-1} + \Delta \Delta^T) = \text{cov}[\hat{\beta}^{OLS}] + \sigma_e^2 \Delta \Delta^T$$

$$\text{since w/ } \text{cov}[\hat{\beta}^{OLS}] = \text{cov}[\hat{\beta}]|_{\Delta=0} = \sigma_e^2 (G^T G)^{-1}$$

→ Denoting $\hat{\sigma}_i$ as the i th diagonal element of $\text{cov}[\hat{\beta}]$, and by construction, the diagonal elements

$$\text{of } \Delta \Delta^T \text{ are also } \geq 0 \rightarrow \text{var}[\hat{\beta}_i] \geq \text{var}[\hat{\beta}_i^{OLS}] \text{ for each } i = 0, \dots, P-1, \text{ w/ equality when } \Delta = 0.$$

- This means the OLS estimator $\hat{\beta}^{OLS}$ is the MVU estimator of θ . Such an estimator is a **BEST Linear Unbiased Estimator (BLUE)**.

- By considering the expression for $\text{cov}[\hat{\beta}]$, we can see that $\hat{\beta}^{OLS}$ is the unique BLUE for the general linear model.

- The only assumption used is that the noise process $\{\epsilon_n\}$ is zero mean and white (we did not assume any probability distribution for $\{\epsilon_n\}$).

- However, if we do know the distribution of $\{\epsilon_n\}$, it is possible that the linear estimator can be beaten by a nonlinear estimation method.

- If in addition, we have prior probability information about θ , $p(\theta)$, then a Bayesian estimator can give even better mean squared performance at the cost of some small bias in the estimate.

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Likelihood-based inference

Error sequence $\mathbf{e}$ and the linear Gaussian model

- We assume the error sequence $\mathbf{e}$ to be drawn from an iid noise distribution w/ pdf $p_e(e)$, so

$$p(\mathbf{e}) = p_e(e_0) p_e(e_1) \dots p_e(e_{N-1})$$

and we also assume the noise process is white (though not necessarily zero mean).

- For the linear Gaussian model (Gauss-Markov model), p_e is the zero-mean Gaussian distribution.

$$p_e(x) = N(x | \mu=0, \sigma^2=\sigma_e^2) = \frac{1}{\sqrt{2\pi}\sigma_e} e^{-\frac{x^2}{2\sigma_e^2}}$$

The error sequence $\mathbf{e}$ thus has a joint pdf $p(\mathbf{e})$, given by

$$p(\mathbf{e}) = \prod_{n=0}^{N-1} N(e_n | \mu=0, \sigma^2=\sigma_e^2) = \prod_{n=0}^{N-1} \frac{1}{\sqrt{2\pi}\sigma_e} e^{-\frac{e_n^2}{2\sigma_e^2}} = \left(\frac{1}{\sqrt{2\pi}\sigma_e}\right)^N e^{-\frac{1}{2\sigma_e^2} \sum_{n=0}^{N-1} e_n^2} = \frac{1}{(2\pi\sigma_e^2)^{N/2}} e^{-\frac{\mathbf{e}^T \mathbf{e}}{2\sigma_e^2}} = N(\mathbf{e} | \mu=0, \mathbf{C}_e=\sigma_e^2 \mathbf{I})$$

- Note the multivariate Gaussian density function for a length N random column vector $\mathbf{x}$, $f_{\mathbf{x}}(\mathbf{x})$ is

$$f_{\mathbf{x}}(\mathbf{x}) = \frac{1}{(2\pi)^{N/2} |\mathbf{C}_{\mathbf{x}}|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \mathbf{C}_{\mathbf{x}}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right)$$

where $\boldsymbol{\mu} = E[\mathbf{x}]$ is the mean vector and $\mathbf{C}_{\mathbf{x}} = E[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^T]$ is the covariance matrix.

(Elementwise, we have $[\mathbf{C}_{\mathbf{x}}]_{ij} = C_{x_i x_j}$, the covariance b/w elements x_i and x_j).

Maximum likelihood (ML) estimator $\hat{\boldsymbol{\theta}}^{ML}$

- The ML estimator treats the parameters $\boldsymbol{\theta}$ as unknown const. dot which we incorporate no prior information.

The observed data $\mathbf{x}$ is considered random and we can often obtain the pdf for $\mathbf{x}$ when $\boldsymbol{\theta}$ is known.

- This pdf is termed the likelihood $L(\mathbf{x}; \boldsymbol{\theta})$, defined as.

$$L(\mathbf{x}; \boldsymbol{\theta}) = p(\mathbf{x} | \boldsymbol{\theta}).$$

- The ML estimate for $\boldsymbol{\theta}$ is then the value of $\boldsymbol{\theta}$ which max the likelihood for given observations $\mathbf{x}$.

$$\hat{\boldsymbol{\theta}}^{ML} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} p(\mathbf{x} | \boldsymbol{\theta}).$$

i.e. the ML soln corresponds to the parameter vector which would have generated $\mathbf{x}$ w/ highest probability.

- We typically max. the log-likelihood function $I(\mathbf{x}; \boldsymbol{\theta}) = \log(L(\mathbf{x}; \boldsymbol{\theta}))$ rather than $L(\mathbf{x}; \boldsymbol{\theta})$ itself for convenience since $\log(\cdot)$ is a monotonically increasing function, the two solns are identical.

- For the linear Gaussian model, $\mathbf{x} = \mathbf{G}\boldsymbol{\theta} + \mathbf{e}$, we do a vector change of variable $\mathbf{e} \rightarrow \mathbf{x}$ to get the likelihood $p(\mathbf{x} | \boldsymbol{\theta})$. (We are conditioning on $\boldsymbol{\theta} \rightarrow$ treat $\mathbf{G}\boldsymbol{\theta}$ as a const. term for change of variable)

$$\frac{\partial x_i}{\partial e_j} = \begin{bmatrix} 1 & \text{if } i=j \\ 0 & \text{o/w} \end{bmatrix} \rightarrow \mathbf{J}^T = \mathbf{I} \rightarrow p(\mathbf{x} | \boldsymbol{\theta}) = p_e(\mathbf{x} - \mathbf{G}\boldsymbol{\theta}) |\mathbf{J}|^{-1} = p_e(\mathbf{x} - \mathbf{G}\boldsymbol{\theta})$$

Expanding this out, we get

$$L(\mathbf{x}; \boldsymbol{\theta}) = p_e(\mathbf{x} - \mathbf{G}\boldsymbol{\theta}) = \frac{1}{(2\pi\sigma_e^2)^{N/2}} \exp\left(-\frac{1}{2\sigma_e^2}(\mathbf{x} - \mathbf{G}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{G}\boldsymbol{\theta})\right)$$

$$\therefore I(\mathbf{x}; \boldsymbol{\theta}) = \log L(\mathbf{x}; \boldsymbol{\theta}) = -\frac{N}{2} \log 2\pi\sigma_e^2 - \frac{1}{2\sigma_e^2}(\mathbf{x} - \mathbf{G}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{G}\boldsymbol{\theta}) = -\frac{N}{2} \log 2\pi\sigma_e^2 - \frac{1}{2\sigma_e^2} \mathbf{e}^T \mathbf{e} = -\frac{N}{2} \log 2\pi\sigma_e^2 - \frac{1}{2\sigma_e^2} \mathbf{e}^T \mathbf{e}$$

- $\rightarrow$ Maximising $I(\mathbf{x}; \boldsymbol{\theta})$ wrt $\boldsymbol{\theta}$ is equivalent to minimising $\mathbf{J}^T \mathbf{e}^T \mathbf{e}$ wrt $\boldsymbol{\theta} \rightarrow$ same as OLS estimation, so

$$\hat{\boldsymbol{\theta}}^{ML} = \hat{\boldsymbol{\theta}}^{OLS} = (\mathbf{G}^T \mathbf{G})^{-1} \mathbf{G}^T \mathbf{x}$$

- The OLS soln is a special case of the ML soln where the error process $\{e_n\}$ is zero-mean, independent and Gaussian w/ fixed variance.

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Estimating the noise variance

- The noise variance σ^2 of the linear Gaussian model can be estimated by ML.
- consider the loglikelihood function of $\theta = \theta^{ML}$, now also considered as a function of σ^2 .

$$I(x; \theta = \theta^{ML}, \sigma^2) = \log L(x; \theta = \theta^{ML}, \sigma^2) = -\frac{N}{2} \log 2\pi\sigma^2 - \frac{1}{2\sigma^2} (x - \theta^{ML})^T (x - \theta^{ML}) = -\frac{N}{2} \log 2\pi\sigma^2 - \frac{J^{ML}}{2\sigma^2}$$

where $J^{ML} = (x - \theta^{ML})^T (x - \theta^{ML})$ is the m.h. squared error term correspondingly to ML optimisation.

Differentiating wrt σ^2 and setting to zero,

$$\frac{\partial I(x; \theta = \theta^{ML}, \sigma^2)}{\partial \sigma^2} = -\frac{N/2}{\sigma^2} + \frac{J^{ML}}{2\sigma^2} = 0 \quad \rightarrow \quad \sigma^2^{ML} = \frac{J^{ML}}{N}$$

i.e. the noise variance σ^2^{ML} is estimated as the mean squared error at the ML parameter set.

Bayesian inference

Bayesian methods

- The ML method treats parameters θ as unknown const. If we instead treat the parameters θ as RVs, we can assign a prior pdf $p(\theta)$. key/diff from likelihood-based / Bayesian inference
- The prior pdf $p(\theta)$ should ideally express some prior knowledge abt. the rel. probability of diff. parameter values θ before the data x are observed. (if nothing abt the parameters θ is known a priori, $p(\theta)$ should not express initial preference for a set of parameters over others).
- The precise form assigned to $p(\theta)$ req. careful consideration since misleading results can be obtained from erroneous priors.
- We can use Bayesian to find the posterior probability $p(\theta|x)$, which can be used to estimate random parameters θ from a random vector x of observations.

$$p(\theta|x) = \frac{p(x|\theta)p(\theta)}{p(x)}$$

(note $p(x|\theta)$ is the likelihood; $p(\theta)$ is the prior; $p(x)$ is the marginal likelihood)

- The generation of the posterior from the prior when data x is observed can be thought of as a refinement to any previous knowledge abt. the parameters θ . ($p(\theta) \xrightarrow{\text{data } x} p(\theta|x)$).
- The marginal likelihood $p(x)$ is useful in model selection problems, but is const. for any given observation $x \rightarrow$ ignored if we only want to optimise the posterior probabilities wrt θ .

$$p(\theta|x) \propto p(x|\theta)p(\theta)$$

- $p(x)$ could be calculated using $p(x) = \int p(x|\theta)p(\theta)d\theta = \int p(x|\theta)p(\theta)d\theta$, and effectively serves as the normalising const. for the posterior density $p(\theta|x)$.
- All the distributions above are implicitly conditioned upon all prior modelling assumptions, so some texts adopt the notation $p(\theta|x, M)$, where M denotes all of the modelling and distributional assumptions that are being made (not just the assumed prior $p(\theta)$).
- LS / ML methods only provide a point estimate of θ , whereas Bayesian methods produce a posterior pdf w/ values defined for all $\theta \rightarrow$ fully interpretate probability distribution

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Maximum a posteriori (MAP) estimator θ^{MAP}

- The MAP estimate for θ is the value of θ which max the posterior for given observations x and prior $p(\theta)$

$$\theta^{MAP} = \underset{\theta}{\operatorname{argmax}} p(\theta|x)$$

i.e. the MAP soln corresponds to the parameter vector θ inferred from x w/ highest probability

- For the linear Gaussian model, suppose the prior $p(\theta)$ is the multivariate Gaussian, i.e.

$$p(\theta) = N(\theta | m_0, \Sigma_0) = \frac{1}{(2\pi)^{r/2} |\Sigma_0|^{1/2}} \exp\left(-\frac{1}{2}(\theta - m_0)^T \Sigma_0^{-1} (\theta - m_0)\right)$$

where r is the no. of parameters in θ .

(Note if we don't know much a priori, we usually select $m_0 = 0$ and $\Sigma_0 = \sigma_0^2 I$)

The likelihood $p(x|\theta)$ takes the same form as before for the ML estimator, so the posterior $p(\theta|x)$ is

$$p(\theta|x) \propto p(\theta) p(x|\theta) = \frac{1}{(2\pi)^{r/2} |\Sigma_0|^{1/2}} \exp\left(-\frac{1}{2}(\theta - m_0)^T \Sigma_0^{-1} (\theta - m_0)\right) \cdot \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2}(x - \theta)^T (x - \theta)\right)$$

Considering $-2\sigma^2 \log(p(\theta|x))$, we have

$$-2\sigma^2 \log(p(\theta|x)) = \sigma^2 (\theta - m_0)^T \Sigma_0^{-1} (\theta - m_0) + (x - \theta)^T (x - \theta) + \text{const.}$$

$$= \sigma^2 (\theta^T \Sigma_0^{-1} \theta + m_0^T \Sigma_0^{-1} m_0 - 2m_0^T \Sigma_0^{-1} \theta) + x^T x + \theta^T \theta - 2x^T \theta + \text{const.}$$

(i) Differentiating wrt θ and setting to zero,

$$\frac{\partial (-2\sigma^2 \log(p(\theta|x)))}{\partial \theta} = \sigma^2 (2\Sigma_0^{-1} \theta + 0 - 2\Sigma_0^{-1} m_0) + 0 + 2\Sigma_0^{-1} \theta - 2\Sigma_0^{-1} x = 0$$

$$\therefore (\sigma^2 \Sigma_0^{-1} + \Sigma_0^{-1}) \theta = \sigma^2 \Sigma_0^{-1} m_0 + \Sigma_0^{-1} x \rightarrow \boxed{\theta^{MAP} = (\Sigma_0^{-1} + \sigma^2 \Sigma_0^{-1})^{-1} (\Sigma_0^{-1} x + \sigma^2 \Sigma_0^{-1} m_0)}$$

(ii) Grouping terms in the form $\theta^T \theta \rightarrow 2\theta^T b$ then completing the square

$$-2\sigma^2 \log(p(\theta|x)) = \theta^T (\Sigma_0^{-1} + \sigma^2 \Sigma_0^{-1}) \theta - 2(x^T \Sigma_0^{-1} + \sigma^2 m_0^T \Sigma_0^{-1}) \theta + \text{const.}$$

$$= \theta^T (\Sigma_0^{-1} + \sigma^2 \Sigma_0^{-1}) \theta - 2(\Sigma_0^{-1} x + \sigma^2 \Sigma_0^{-1} m_0)^T \theta + \text{const.}$$

$$= \theta^T \theta - 2b^T x + \text{const.} = (\theta - b)^T (\theta - b) + \text{const.}$$

$$\therefore \theta^{MAP} = \underline{b}^T b, \text{ where } \underline{b} = (\Sigma_0^{-1} + \sigma^2 \Sigma_0^{-1}), \quad b = (\Sigma_0^{-1} x + \sigma^2 \Sigma_0^{-1} m_0)$$

(Note we assumed Σ_0^{-1} is invertible, which is the case if Σ_0 is full rank)

- By completing the square, for $-2\sigma^2 \log(p(\theta|x))$, and considering the exponent of $p(\theta|x)$,

$$-2\sigma^2 \log(p(\theta|x)) = (\theta - \theta^{MAP})^T \underline{b} (\theta - \theta^{MAP}) + \text{const.}$$

$$\therefore \text{Exponent of } p(\theta|x) = -\frac{1}{2\sigma^2} (\theta - \theta^{MAP})^T \underline{b} (\theta - \theta^{MAP}) \rightarrow \text{multivariate Gaussian } N(m_0^{POST}, \Sigma_0^{POST}),$$

$$m_0^{POST} = \theta^{MAP}, \quad \Sigma_0^{POST} = \left(\frac{\underline{b}}{\sigma^2}\right)^{-1} = \sigma^2 \underline{b}^{-1}$$

since the remaining terms do not depend on θ , on the multivariate density function must be proper

we can conclude the posterior $p(\theta|x)$ is itself a multivariate Gaussian

$$\boxed{p(\theta|x) = N(\theta | \theta^{MAP}, \sigma^2 \underline{b}^{-1})}$$

- comparing $\theta^{MAP} = (\Sigma_0^{-1} + \sigma^2 \Sigma_0^{-1})^{-1} (\Sigma_0^{-1} x + \sigma^2 \Sigma_0^{-1} m_0)$ w/ $\theta^{ML} = (\Sigma_0^{-1})^{-1} \Sigma_0^{-1} x$, we can see in the limit $\Sigma_0^{-1} = 0$, $\theta^{MAP} = \theta^{ML}$

($\Sigma_0^{-1} \rightarrow 0 \rightarrow \sigma_0^2 \rightarrow \infty$, i.e. the prior tends to be flat/uniform (equitable θ) $\rightarrow$ impose no prior information)

- The ML soln is a special case of the MAP soln, where a uniform prior is assigned to θ .

- The MAP estimate also tends to the ML estimate when the likelihood is sharply peaked around its max, compared w/ the prior (i.e. $\sigma^2 \ll \sigma_0^2$). This is the case for large N .

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Minimum mean-squared error (MMSE) estimator. θ^{MMSE}

- consider a cost function $c(\hat{\theta}, \theta)$ which expresses the cost of estimating the parameter as $\hat{\theta}$ when the true value is θ . ($c(\hat{\theta}, \theta)$ should be nonnegative and usually satisfied $c(\hat{\theta}, \theta) = 0$).
- the form of the cost function $c(\hat{\theta}, \theta)$ depends on the req. of the problem. A cost of 0 indicates that the estimate is perfect for our req. (not necessarily $\hat{\theta} = \theta$), while the costs indicate poorer estimates.
- The expected cost over all of the unknown parameters, conditioned on the observed data x is

$$E[c(\hat{\theta}, \theta) | x=x] = \int c(\hat{\theta}, \theta) p(\theta | x) d\theta$$

- The MMSE estimate for θ is the value of $\hat{\theta}$ which min. the average squared error for given data x .

$$\theta^{MMSE} = \arg \min_{\hat{\theta}} E[(\hat{\theta} - \theta)^2 | x=x]$$

(Note the cost function $c(\hat{\theta}, \theta)$ is $c(\hat{\theta}, \theta) = (\hat{\theta} - \theta)^2$)

Any small bias introduced can be tolerated provided the MSE is low.

- In general, the MSE can be expressed as

$$J = E[(\hat{\theta} - \theta)^2 | x=x] = \int (\hat{\theta} - \theta)^2 p(\theta | x) d\theta$$

Differentiating wrt $\hat{\theta}$ and setting to zero,

$$\frac{dJ}{d\hat{\theta}} = \frac{d}{d\hat{\theta}} \int (\hat{\theta} - \theta)^2 p(\theta | x) d\theta = \int \frac{d}{d\hat{\theta}} (\hat{\theta} - \theta)^2 p(\theta | x) d\theta = \int 2(\hat{\theta} - \theta) p(\theta | x) d\theta = 0$$

$$\therefore \int \hat{\theta} p(\theta | x) d\theta = \int \theta p(\theta | x) d\theta \rightarrow \theta^{MMSE} = E[\theta | x=x] = \int \theta p(\theta | x) d\theta$$

since $\int \hat{\theta} p(\theta | x) d\theta = \hat{\theta} \int p(\theta | x) d\theta = \hat{\theta}$

- For the linear Gaussian model, the posterior distribution is $p(\theta | x) = N(\theta | \hat{\theta}^{MAP}, \sigma_e^2 \hat{\Sigma}^{-1})$, so

$$\theta^{MMSE} = E[\theta | x=x] = \theta^{MAP} = (\hat{\Sigma}^T \hat{\Sigma} + \sigma_e^2 \hat{\Sigma}^{-1})^{-1} (\hat{\Sigma}^T x + \sigma_e^2 \hat{\Sigma}^{-1} m_0)$$

Summary of estimators

① Least squares (θ^{LS})

- ✓: Req. no knowledge of probability distributions
- ✗: cannot incorporate prior knowledge of $p(\theta)$
- ✓: Usually simplest to implement
- ✓: Guarantee of performance as BLUE estimator (not necessarily better than nonlinear estimators)

② Maximum likelihood (θ^{ML})

- ✓: Guarantee of performance when N is large
- ✗: Req. knowledge of noise probability distribution
- ✗: Cannot incorporate prior knowledge of $p(\theta)$
- ✗: Harder to implement than LS for non-Gaussian case.

③ Bayesian ($\theta^{MAP}, \theta^{MMSE}$)

- ✓: Incorporates prior knowledge of $p(\theta)$
- ✗: Req. knowledge of noise probability distribution
- ✓: Guarantee of performance for any N (given correct $p(\theta)$)
- ✗: Req. knowledge of prior distribution $p(\theta)$
- ✗: Can be harder to implement than LS/ML